Framework and toolkit for infection prevention and control outbreak preparedness, readiness and response

Implementation manual

World Health Organization

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Abbreviations

CDC	US Centers for Disease Control and Prevention
CDI	*Clostridioides difficile* infection
COVID-19	Coronavirus Disease 2019
DRASA	Dr Ameyo Stella Adadevoh Health Trust
EBOD	Ebola disease
ECHO	Extension for Community Healthcare Outcomes
HAI	Healthcare-associated infection
HeRams	Health resources and services availability monitoring systems
HF	Health Facility
IMS	Incident management system
IPC	Infection prevention and control
IPCAF	Infection prevention and control assessment tool for health facilities
IPCAT	Infection prevention and control assessment tool for the national level
IPC-OFT	Framework and toolkit for infection prevention and control outbreak preparedness, readiness, and response at the national level
MERS-CoV	Middle east respiratory syndrome coronavirus
MMS	Multimodal strategy
NCDC	Nigeria Centre for Disease Control and Prevention
NGO	Nongovernmental organization
PHE	Public health emergencies
PPE	Personal protective equipment
PTF	Presidential task force
RCCE	Risk communication and community engagement
SOP	Standard operating procedure
ToR	Terms of reference
UNICEF	United Nations Children's Fund
USAID	United States Agency for International Development
WASH	Water, sanitation and hygiene
WHO	World Health Organization

Glossary

Action plan: a detailed set of instructions to follow in order to solve a problem or achieve something *(2)*.

Administrative controls: established work practices that reduce the duration, frequency, or intensity of exposure to hazards (e.g. work-process training; job rotation; ensuring adequate rest breaks; limiting access to hazardous areas or machinery; adjusting line speeds *(3)*.

Budget: a plan to show how much money a person or organization will earn and how much they will need or be able to spend *(2)*.

Care worker: provider of direct personal care services in the home, in health care and residential settings, assisting with routine tasks of daily life and performing a variety of other tasks of a simple and routine nature *(4)*.

Contingency planning: part of cycle in which the identification and regular monitoring of risks, vulnerabilities and capacities informs the plan and implementation of measures to mitigate the risks and prepare to respond *(5)*.

Drill: a coordinated, supervised exercise, normally carried out in a repeated fashion to test or train a single specific operation or function. A drill aims to practice and perfect one small part of a response plan, and should be as realistic as possible, employing any equipment or apparatus necessary for that part *(6)*.

Engineering controls: measures taken to reduce or prevent hazards from coming into contact with workers (e.g. by modifying equipment or the workspace, using protective barriers, ventilation) *(3)*.

Epidemic: the occurrence of a number of cases of a disease that is unusually large or unexpected for a given place and time *(8)*.

Health-care-associated infection (also referred to as "nosocomial" or "hospital infection"): an infection occurring in a patient during the process of care in a hospital or other health facility, which was not present or incubating at the time of admission. Health care-associated infections can also appear after discharge. They represent the most frequent adverse event associated with patient care *(11)*.

Health emergency: a type of event or imminent threat that produces or has the potential to produce a range of health consequences, and which requires coordinated action, urgent and often non-routine *(8)*.

Health and care worker: anyone primarily engaged in actions with the primary intent of enhancing health *(4)*.

Incident management system (IMS): the standardized structure and approach that WHO has adopted to manage its response to public-health events and emergencies, and to ensure that the Organization follows best practices in emergency management. WHO's Incident Management System comprises six critical functions: Leadership, Partner Coordination, Information and Planning, Health Operations and Technical Expertise, Operations Support and Logistics, and Finance and Administration. *(8)* The IPC is a pillar included in the Health Operations and Technical Expertise function.

Infection prevention and control professional (IPCP): any health care professional (medical doctor, nurse or other health-related professional) who has completed a certified postgraduate IPC-training course or a nationally or internationally recognized postgraduate course on IPC, or another core discipline that includes IPC as a core part of the curriculum as well as IPC practical and clinical training *(9)*.

Infection prevention and control (IPC) minimum requirements: IPC standards that should be in place at both national and health facility level to provide minimum protection and safety to patients, health care workers and visitors, based on the WHO core components for IPC programmes. The existence of these requirements constitutes the starting point for building additional critical elements of the IPC core components according to a stepwise approach based on assessments of the local situation *(10)*.

Infection prevention and control (IPC) national programme: the ensemble of policies, goals, strategies, legal, technical framework and monitoring of health care-associated infections *(11)*.

IPC committee: a multidisciplinary group composed of interested stakeholders from across the facility who interact with and advise the IPC team. At national level, an official multidisciplinary group, committee or equivalent structure should be established to interact with the IPC technical team *(11)*.

IPC focal point: an IPC professional (as defined above) appointed to be in charge of IPC at the national, subnational or facility/organization level *(11)*.

Outbreak: often used synonymously with "epidemic", usually to indicate localize as opposed to generalized epidemics *(8)*.

Preparedness: the knowledge and capacities developed by governments, response and recovery organizations, communities and individuals to effectively anticipate, respond to and recover from the impacts of likely, imminent or current disasters *(8)*.

Readiness (synonym "operational readiness"): the ability to quickly and appropriately respond when required *(8)*.

Response: the provision of emergency services and public assistance during or immediately after disasters to save lives; reduce health impacts; ensure public safety; and meet the basic subsistence needs of the people affected *(8)*.

Standard operating procedure (SOP): set of step-by-step instructions compiled by an organization to help workers carry out routine operations in the most effective manner possible *(10)*.

Tabletop exercise: a facilitated discussion of an emergency situation, generally in an informal, low-stress environment. It is designed to elicit constructive discussion among participants; to identify and resolve problems; and to refine existing operation plans *(12)*.

Water, sanitation and hygiene (WASH) in health facilities: refers to the provision of water, sanitation, health-care waste management, hygiene and environmental cleaning, infrastructure and services across all parts of a facility *(7)*.

Executive summary

As observed in recent large-scale outbreaks, health facilities (HFs) with limited or no infection prevention and control (IPC) capacities can amplify emerging infectious disease transmission within facilities and communities and across borders. To mitigate the risk of disease transmission, WHO has developed two complementary tools:

1. *The framework and toolkit for infection prevention and control outbreak preparedness, readiness and response at the national level* (IPC-OFT)*(1)*;
2. *The implementation manual for the framework and toolkit for infection prevention and control outbreak preparedness, readiness, and response at the national level* (presented here).

The first tool provides national and subnational authorities with a practical framework of actions for strengthening IPC outbreak preparation, readiness and response, and a toolkit that provides resources to assist in the development of contingency or action plans to achieve this aim.

The second is an actionable tool specifically designed to enhance the uptake of the IPC-OFT. It places more emphasis on practical implementation, aiming to support the countries in effectively applying the principles outlined in the IPC-OFT. This manual has been developed based on the principles of implementation science and lessons learned from previous responses.

Each chapter can be read independently, according to the emergency phase and user need. Chapter 1 introduces implementation science concepts, including potential barriers and suggestions on how to overcome them. The strategy for the implementation includes the multimodal strategy, a core component of infection prevention and control programmes at national level.

Chapters 2, 3 and 4 provide a step-by-step approach according to the health emergency phases, respectively preparedness, readiness and response. Each chapter introduces key actions according to their priority level (immediate, early, advanced) and refers to tools and resources that can be used at each step.

Chapter 5 provides country case studies to illustrate key actions for end users to adapt to their own local needs.

A list of tools and resources is provided to support the implementation journey and documentation of the process.

Chapter 1

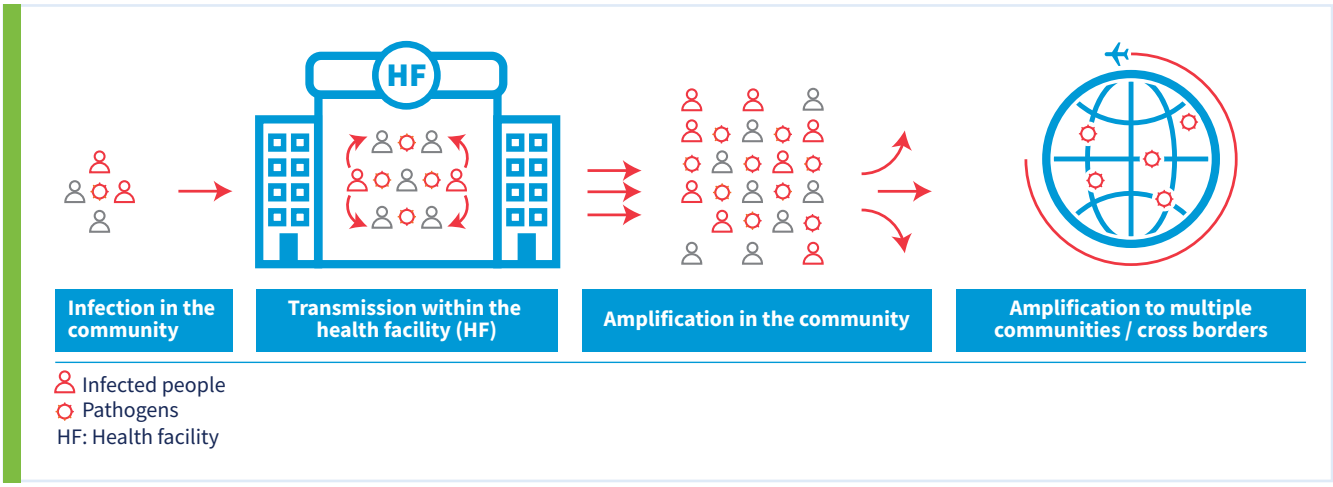
Introduction to the implementation manual

Chapter 1. Introduction to the implementation manual

1.1 Background

Infectious disease epidemics and pandemics pose a significant risk to global health, security and socioeconomic stability *(13)*. From pandemic or seasonal influenza to Ebola Disease (EBOD), Middle East-respiratory syndrome coronavirus (MERS-CoV), and Coronavirus disease 2019 (COVID-19), outbreaks caused by epidemic- or pandemic-prone pathogens are increasing in frequency, scale and impact. As observed in previous large-scale outbreaks, health facilities (HFs) with limited or no infection prevention and control (IPC) capacities can amplify infectious disease transmission within the HFs themselves, within communities and possibly across borders (Figure 1.1).

Figure 1.1. Potential community amplification of an infectious disease outbreak by health facilities.



Evidence-based IPC measures in HFs are critical for preventing and containing outbreaks and ensuring the ongoing delivery of safe, effective and quality health care. Strengthening IPC preparedness and operational readiness are pivotal for a prompt and effective outbreak response. This includes investing in IPC at the national and HF levels to mitigate the risk of health care-associated transmission of disease to patients or health and care workers, and contributing to the timely containment of the outbreak. The *Framework and toolkit for infection prevention and control outbreak preparedness, readiness and response at the national level* (IPC-OFT) *(1)* provides national and subnational authorities with:

1. a practical framework of actions for strengthening IPC outbreak preparation, readiness and response; and
2. a toolkit that provides resources to assist in the development of contingency or action plans to strengthen IPC outbreak preparedness, readiness and response.

Various strategies can be used to implement evidence-based practices. Scientific evidence supports the use of multimodal versus unimodal strategies as highly effective with respect to IPC outcomes *(11)*. The multimodal strategy (MMS) is a core component of effective IPC programmes at local and national levels *(14)*. A MMS consists of several elements (three or more, usually five) implemented in an integrated way to guide action and provide a clear focus for the implementer (Box 1.1).

Additional information on the multimodal strategy can be found at Interim practical manual supporting national implementation of the who guidelines on core components of infection prevention and control programmes.

Box 1.1. Elements of multimodal strategy for improvement of infection prevention and control, according to the World Health Organization *(11).*

Build it
System change: availability of the appropriate resources, including infrastructure and supplies needed to enable IPC practices.

Teach it
Training and education: of health care workers and key players (e.g. managers, environmental services, politicians).

Check it
Monitoring and feedback: this encompasses activities undertaken to assess and document the problem and provide feedback to relevant stakeholders on practice improvement.

Live it
Culture change: actions to strengthen the ready-to-respond climate and learning process from each outbreak response experience.

Sell it
Reminders and communications: this includes the communications plans and reminders in the workplace.

To facilitate implementation of *IPC-OFT* this manual has been developed based on MMS and evidence-based best practices. It provides a step-by-step guide and tools to be used or adapted according to national or subnational context.

1.2 Objective

To provide national and subnational authorities with step-by-step directions and a tool set to implement the *IPC-OFT.*

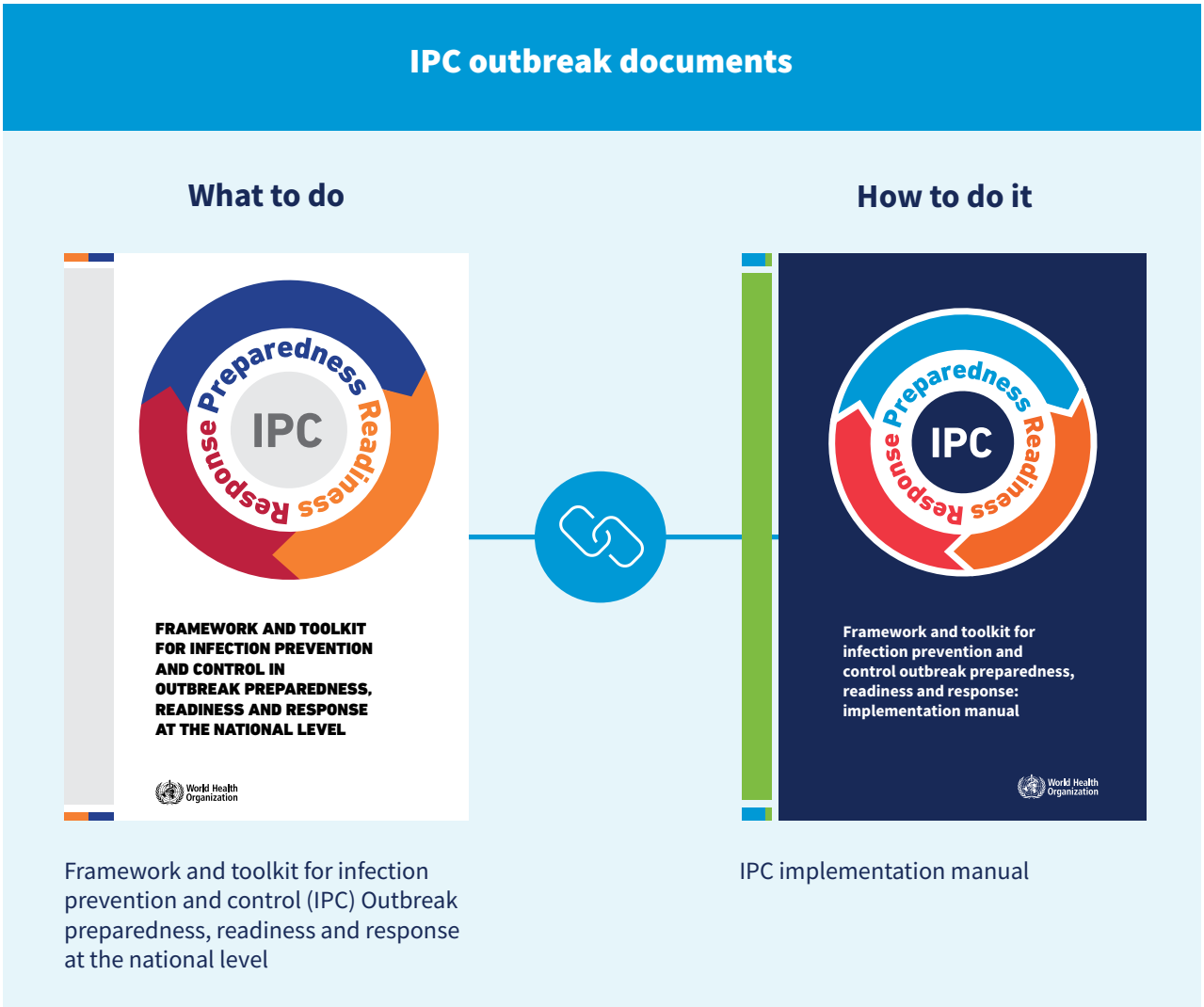
Target audience

- National and subnational IPC focal points and decision-makers, outbreak response focal points (IPC task force, IPC focal point and incident managers).
- Other target audiences include public health institutions, disaster management agencies, safety and quality leads and managers, regulatory bodies, and allied organizations, including academia, national IPC professional bodies, and nongovernmental organizations involved in IPC activity and emergency response.

1.3 How to use this implementation manual

This implementation manual accompanies the IPC-OFT *(1)* (Figure 1.2), which contains the core information on IPC outbreak preparedness, readiness and response.

Figure 1.2 The use of this implementation manual jointly with the Framework and Toolkit for Infection Prevention and Control Outbreak Preparedness, Readiness and Response at the national level *(1)*



Chapters 2, 3 and 4 provide guidance on how to perform **key actions** according to each **emergency management phase**, and detail **priorities**. A summary of elements on these chapters is available in Table 1.1.

Assessment tools, templates, checklists and prototypes are also provided in these chapters that can be used according to the needs. Some have identified tools needed to support implementation that have not yet been developed; WHO in collaboration with partners will identify mechanisms to address these gaps.

By answering **two targeted questions** (Box 1.2 and Box 1.3), you will be directed to the appropriate chapter to support implementation, depending on the current country emergency phase.

Table 1.1 Summary of phases, priorities and key actions for infection prevention and control in emergencies at national level and corresponding multimodal strategy elements

Emergency management phase	Priority	Key actions	Multimodal element
Preparedness	Immediate	Evaluate IPC capacity	Check it
	"	Develop national IPC preparedness and response plan	Build it
	"	Develop national IPC outbreak guidelines	Build it
	"	Develop national IPC outbreak training programme	Teach it
	"	Coordinate with national surveillance and reporting programme	Check it
	"	Develop national or subnational IPC communication strategy	Sell it
	"	Develop surge capacity plans for IPC resources	Build it
	Early	Establish IPC outbreak taskforce	Build it
	"	Establish IPC partner mapping and networking	Build it
	Advanced	Assess IPC outbreak preparedness	Check it
Readiness	Immediate	Establish IPC outbreak taskforce	Build it
	"	Ready the national IPC outbreak response plan	Build it
	"	Adapt surge capacity plans for IPC resources	Build it
	"	Adapt national IPC outbreak training programme	Teach it
	"	Coordinate with national surveillance and reporting programme	Check it
	"	Adapt national or subnational IPC communication strategy	Sell it
	Early	Prepare IPC partners	Live it
	"	Prepare IPC stakeholder networks	Live it
	Advanced	Assess IPC outbreak readiness	Check it
Response	Immediate	Activate IPC outbreak task force	Build it
	"	Activate national IPC outbreak response plan	Build it
	"	Activate surge capacity plans for IPC resources	Build it
	"	Leverage national IPC outbreak training programme	Teach it
	"	Activate national surveillance and reporting programme	Check it
	"	Activate national or subnational IPC communication strategy	Sell it
	Early	Monitor for resurgence of cases	Check it
	Advanced	Assess IPC outbreak response	Check it

1.4 How to identify which emergency phase the country is in: preparedness, readiness or response

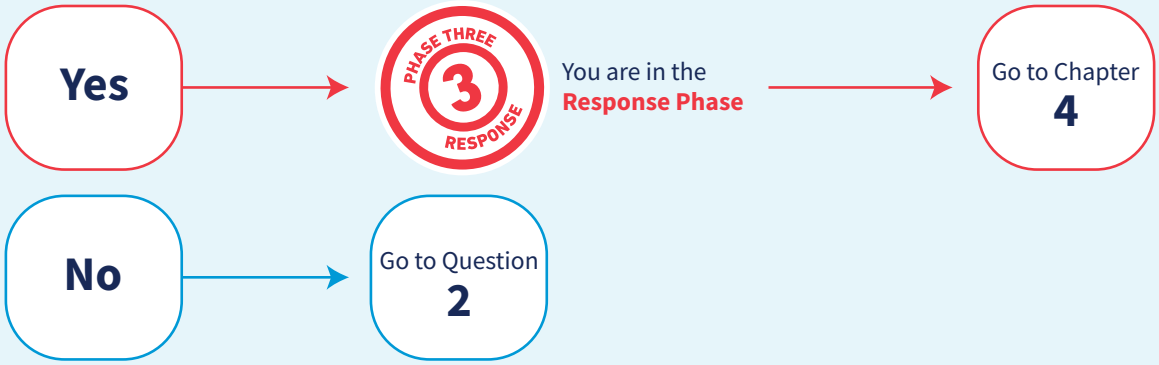
The first step is to identify local disease threats (EBOD, COVID-19, Lassa fever, multi-drug resistant organisms, influenza, etc.); then review available local, regional and global epidemiological data; and answer question 1 and 2 (Box 1.2 and Box 1.3).

Box 1.2. Question 1: identification of country public health emergency management phase at a given moment

Question 1

Do you have **any of the situations** below regarding a particular infectious disease?

- Facing an outbreak of a particular infectious disease?
- Greater than expected incidence of infection compared to the usual background rate for the location (your country, district, a particular location, facility or ward), time and population?
- A single case of certain rare or epidemic- or pandemic-prone diseases?
- A suspected, anticipated or actual event involving microbial contamination of food or water?
- An increased number of cases of highly transmissible disease?



Box 1.3. Question 2: identification of country public health emergency management phase at a given moment

Question 2

Are you facing any of the situations below regarding a particular infectious disease that may lead to an imminent risk of an outbreak?

- A single case of a rare, epidemic or pandemic-prone disease in a neighbouring country, district, or province or a country, district, or province with economic or cultural links?
- An increase in the number of cases of a highly transmissible disease in a neighbouring country, district, or province or a country, district or province with economic or cultural links?
- Has there been a global alert regarding a specific event or disease of national or international public health concern?

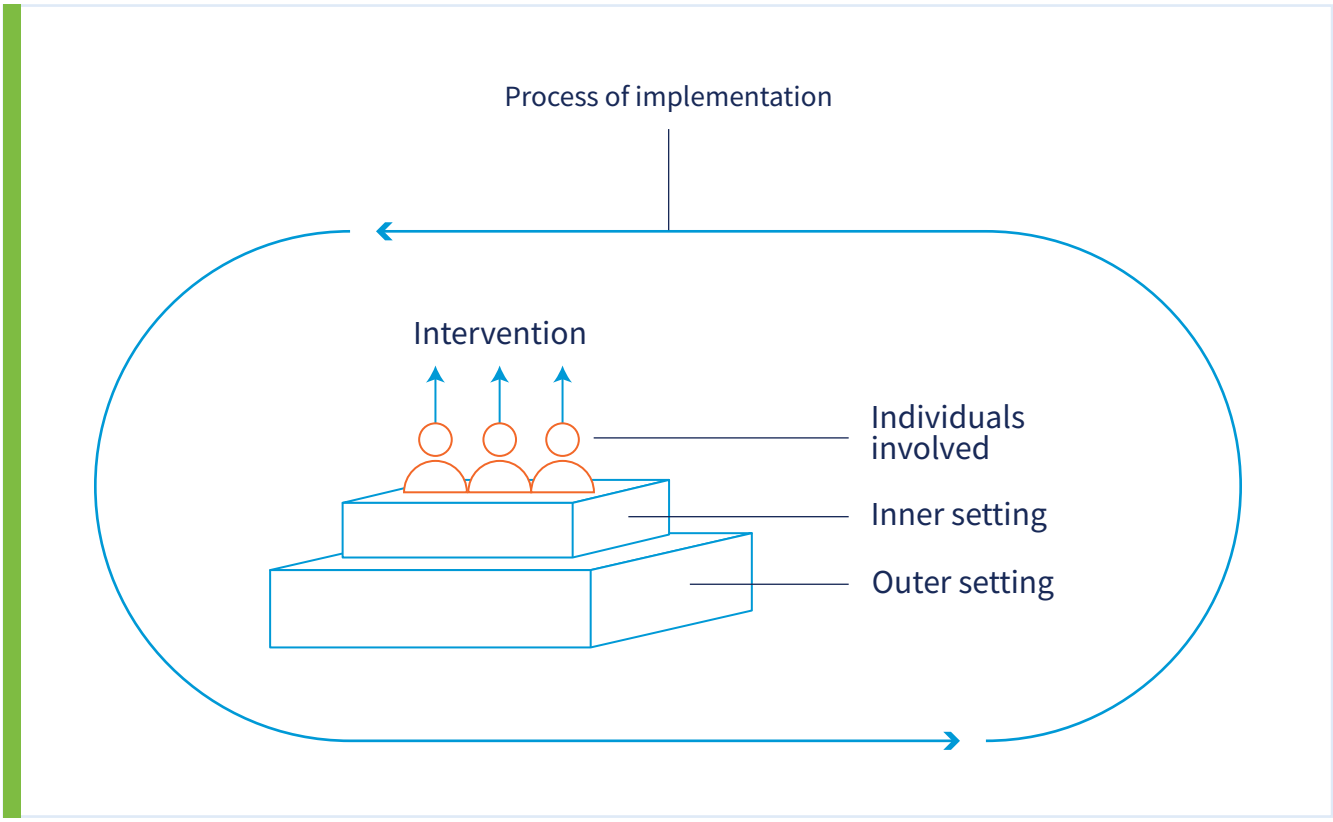
1.5 Navigating the implementation journey

The goal of an implementation process is to improve the adoption, appropriate adaptation, delivery and sustainment of effective interventions by health and care workers, organizations, communities and systems of care *(15, 16)*. Implementation is a practical, proactive and structured process that occurs in the real world.

The context within which an implementation process occurs affects the likelihood of success of any intervention. Five dimensions *(17)* should be considered when developing strategies to achieve maximum IPC outbreak preparedness, readiness and response capacity (Figure 1.3):

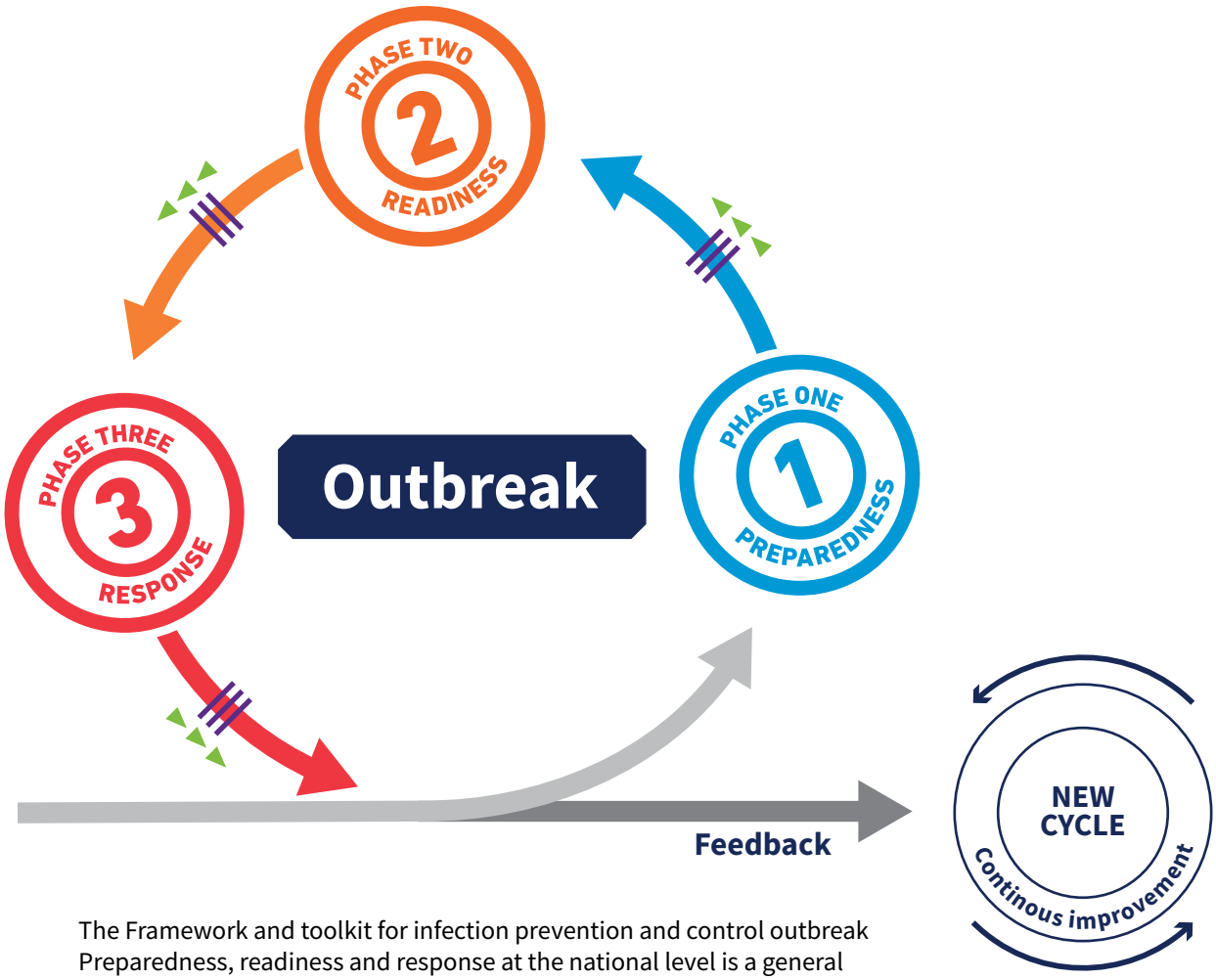
- The *inner setting* is the context within any organization where the intervention is implemented. That context includes culture, structure and networks. In the context of IPC-OFT implementation, this domain can be both the HFs context as well the IPC national programme context.
- The *outer setting* refers to the economic, social, organizational and political context outside of the organization. In the context of IPC-OFT, this domain refers to the national-level environment as well the global context of epidemics/pandemics.
- The process for implementation refers to the methods used to facilitate the adoption of the interventions at all levels. In the context of IPC-OFT, this domain refers to the process of implementation, including the planning, coordination, execution and evaluation.
- The *individuals involved* are those who have a role to play in the implementation process (health and care workers, managers, policymakers, beneficiaries). In the context of IPC-OFT, this domain refers to the stakeholders who will be involved in the process.
- Finally, the *intervention* is any action taken to introduce evidence-based practices. In the context of the IPC-OFT implementation, this refers to the set of key actions proposed.

Figure 1.3. The context and dimensions of implementation that should be considered when implementing the *Framework and toolkit for infection prevention and control outbreak preparedness, readiness and response at the national level*



The journey to implement any new or improved practice requires a continuous improvement process, as there will be barriers, setbacks, enablers and accelerators in the implementation journey (Figure 1.4).

Figure 1.4 Representation of the implementation journey, with barriers and enablers in the continuous improvement of infection prevention and control in outbreak preparedness, readiness and response



The Framework and toolkit for infection prevention and control outbreak Preparedness, readiness and response at the national level is a general framework and methodology, flexible and adaptable to local contexts.

These phases represent a continuous cycle that includes a feedback element.

Implementation barriers are variable and can occur at any point.

Accelerators.

A systematic approach to recognizing, understanding and addressing these barriers and strengthening enablers is essential to selecting the optimal implementation strategies.

Country examples and case studies are provided considering that national or subnational teams can benefit from others' experience. Refer to these when you identify barriers to see if previously identified solutions and strategies might work in your context. Examples of potential barriers and strategies to overcome those related to IPC outbreak preparedness, readiness and response are provided in this document and can be used as a reference (Table 1.2).

Table 1.2. Potential barriers and strategies to overcome barriers related to infection prevention and control in outbreak preparedness, readiness and response

Potential barriers	Strategies to overcome barriers
Lack of awareness of public health emergency terminology: because most IPC focal points do not work exclusively on Public health emergencies (PHE), the language of emergency response is not familiar to most professionals working in IPC	**Build it –** Develop simplified resources with key information **Build it –** Build a good coordination between IPC and NGO partners for response **Sell it –** Develop an awareness campaign and IPC advocacy related to outbreak readiness and response focus on IPC leadership awareness
Lack of IPC coordination: coordination of IPC at different levels can be challenging, with different actors driving IPC within the national and subnational levels; internal national competition for ownership and recognition; and internal competition at national level for priorities	**Build it –** Develop a coordination body that identifies key people who should be engaged in the communication chain use drills or tabletop exercises to help people involved get to know each other and understand the language used for the activities **Live it –** Develop a strategy for stakeholder mapping for IPC during PHEs and engage stakeholders from the beginning of the plan **Live it –** Develop strategies to reduce internal competition by acknowledging the contribution(s) of each department/stakeholder and their autonomy
Lack of support for HFs from the national level: this is a major barrier in many HFs, lack of dedicated personnel, budget, material provision and water	**Build it –** Coordinate with NGO partners to support implementation of essential IPC structure in HFs **Check it –** Use HF assessment tools to systematically identify priority gaps in IPC at HF level (e.g. IPCAT*(18)*, WASH FIT*(19)*) **Live it –** Strengthen the links between IPC and WASH teams
Lack of support for HFs from the national level: lack of funding at national level and district level to support HFs; lack of knowledge of IPC; lack of IPC focal point at national level; other national competing priorities; lack of understanding about the roles of IPC programmes; lack of dedicated budgets	**Build it –** Integrate IPC in the efforts to strengthen the health system as a core part of the response **Build it –** Focus on the development of the IPC national programme, including the outbreak management component, with allocated dedicated budget **Sell it –** Develop an IPC advocacy strategy related to outbreak readiness and response
Lack of IPC specialists: IPC focal points hired for the short term, only primarily for outbreak management or specific projects; lack of understanding and expertise in managing and responding to outbreaks	**Build it –** Develop simple step-by-step protocols with the people who may be involved in the actions this allows those involved to discuss content and solve many issues in advance **Teach it –** Define a basic, standardized IPC training programme for delivery to frontline personnel and coordinate it with partners (e.g. NGO include IPC-trained personnel in the rapid response teams **Teach it –** Develop basic IPC training programmes for supervisors/ administrators/financing people to facilitate their coordination with stakeholders in other teams
Inefficient provision of supplies: frequent shortages of essential IPC items during outbreaks; gaps to establish proper stockpiles; difficulty including newly developed PPE during outbreaks	**Build it –** Map resources in advance to identify any critical gaps and possibilities for resource allocations, including donation channels **Build it –** Develop broad calculations based on the number of HFs and health and care workers to estimate supply needs **Build it –** Establish early communication with logistics personnel to provide specifications and amounts needed of essential IPC items develop a national platform where the HFs can note their needs for IPC supplies or develop a system to keep track of available supplies **Build it –** Develop a rapid but effective local accreditation and regulatory mechanism to assure safe production of IPC logistics, including PPE and hand hygiene products

Potential barriers	Strategies to overcome barriers
Inadequate communication with communities: individuals in the community do not know essential information regarding IPC and become misinformed during outbreaks	**Live it –** Establish early coordination with the risk communication and community engagement team; **Sell it –** Develop "lay" communication materials to spread key message and to promote positive changes in attitudes and behaviours, thereby enabling decision-makers to make healthier choices; **Sell it –** Involve members of the community in the development of key messages and value the expertise and experience across partners; these may include the health workers as a specific community to be targeted;
Gaps in the surveillance system related to IPC-relevant information: during outbreaks critical information such as infections among health and care workers is not timely available	**Build it –** Establish early coordination with the surveillance team to define a process to obtain IPC relevant information during outbreaks; **Check it –** Contribute to the surveillance team investigation by deploying an IPC specialist or team if the nosocomial outbreak is severe or cases are spreading among several HFs or regions, and perform data analyses, providing rapid feedback on trends and support to HFs; **Live it –** Work in cooperation with rapid response teams to get timely information.

Note: PHE – public health emergencies; IPC: infection prevention and control; HF: health facilities; NGO: nongovernmental organization; IPCAT: infection prevention and control assessment tool at the facility level

Chapter 2

Phase 1: IPC outbreak preparedness

Chapter 2. Phase 1: IPC outbreak preparedness

2.1 Introduction

If the answers to questions 1 and 2 in Chapter 1 were "no", the country is in the preparedness phase.

The emergency preparedness phase is the stage for developing public health emergency response plans for relevant hazards, in the absence of an evident threat. The activities involved in the preparedness may take approximately six months to two years before a country or organization is fully prepared to respond to an emerging infectious disease or public health threat *(1)*.

A summary of key priority actions is provided in the Table 2.1.

Table 2.1. Summary of key actions according to the IPC outbreak preparedness phases, priorities and respective multimodal strategy element

Emergency management phase	Priority	Key actions	Multimodal element
Preparedness	Immediate	Evaluate IPC capacity	Check it
	"	Develop national IPC preparedness and response plan	Build it
	"	Develop national IPC outbreak guidelines	Build it
	"	Develop national IPC outbreak training programme	Teach it
	"	Coordinate with national surveillance and reporting programme	Check it
	"	Develop national or subnational IPC communication strategy	Sell it
	"	Develop surge capacity plans for IPC resources	Build it
	Early	Establish IPC outbreak taskforce	Build it
	"	Establish IPC partner mapping and networking	Live it
	Advanced	Assess IPC outbreak preparedness	Check it

2.2 Step by step of key actions and available tools for IPC preparedness

A. Immediate priority: develop an IPC foundation

The immediate actions are the first set of actions required in the public health management phase. The focus is on operationalizing key activities, outline roles and responsibilities, and identifying the necessary resources.

Table 2.2. Key actions, steps, how to do and list of tools resources for the preparedness public health emergency phase immediate priority

Preparedness			
Immediate priority: develop an IPC foundation			
Key action	**Steps**	**How to do**	**List of tools/resources**
Evaluate IPC capacity at national and priority HF levels	Define the work team.	• Identify a group of individuals who have time allocated (at least part time) to participate in the implementation process. • If a national committee exists, link it to the team for IPC outbreak. • Appoint a leader of the process.	
	Develop a preliminary action plan.	• Describe the steps to be taken, by whom and when. This step aims to plan the baseline assessment.	• A_IPC outbreak action plan template
	Perform a baseline assessment of the national IPC programme, if none was performed in the last year.	• If the national IPC programme is in the early stage, use the *Assessment tool of the minimum requirements for IPC programmes at national level.* • If the national IPC programme is more developed, use the *National Infection Prevention and Control Assessment Tool 2 (IPCAT-2).* • Additionally, use the IPC component of the International Health Regulation State Party Self-Assessment Annual Report reporting tool *(IPC indicators C9.1, C9.2 and C9.3 or Joint External Evaluation Tool R4.1, R4.2, R4.3 and technical questions)*.	• Assessment tool of the minimum requirements for IPC programmes at national level • IPCAT-2 • SPAR
	Identify priority gaps in the IPC national programmes regarding outbreak preparedness.	• Once the gaps are identified, a *prioritization tool* can be used. If a large number of stakeholders are involved, use a *prototype of workshop* to set priorities. For a comprehensive prioritization use Strategic toolkit for assessing risks: a comprehensive toolkit for all-hazards health emergency risk assessmen.	• JEE • B_IPC outbreak prioritization template • C_IPC outbreak prioritization workshop prototype
	Develop an action plan based on the results of prioritization process.	• Describe the first steps to be taken, by whom and when.	• A_IPC outbreak action plan template • STAR
	Incorporate outbreak management as a component of the national IPC programme.	• Ensure the outbreak management component of the IPC programme is linked to other relevant programmes (e.g. epidemiology, immunization). • Identify and ensure a dedicated IPC budget is available with provisions for outbreak functions (additional staffing, site visits, etc.).	
	Ensure that the IPC HF level requirements have been assessed and communicated at the HF level.	• Define a strategy to assess critical/reference HFs or a sample of HFs in the country. • Use the WHO infection prevention and control assessment framework (IPCAF) at the facility level. • Additionally, the WASH FIT assessment tool can be used. • If considered relevant, assess IPC in primary care.	• IPCAF • WASH-FIT • Assessment tool on infection prevention and control minimum requirements for primary health care facilities

Key action	Steps	How to do	List of tools/resources
Develop or update the national preparedness and response plan	Map stakeholders for the development or update of the national emergencies preparedness plan.	• The team can use the template for *stakeholders mapping.* • Develop a stakeholder engagement strategy.	• D_IPC outbreak stakeholders mapping template
	Develop or update the national preparedness and response plan.	• Establish a coordination strategy and minimize duplication of efforts. Use the template for *stakeholders mapping*. • Develop a stakeholder engagement strategy.	
	Evaluate/develop the IPC component of any national or subnational emergencies preparedness plan.	• Develop an outbreak management plan with clearly defined goals and objectives. • The WHO guidance for contingency planning can be used as a reference.	• E_IPC outbreak management plan template • WHO guidance for contingency planning
	Develop standard operating procedures (SOPs) according to the preparedness plan.	• Develop priority SOPs (e.g. rapid identification and isolation of suspected cases among patients and workers; safe process for sample collection, transport and lab analysis; rapid contact tracing in the HF and management of contacts).	• F_IPC outbreak SOP template
	Assess requirements for personal protective equipment (PPE) and supplies.	• Identify the number of HFs and respective number of beds to be supplied (e.g. reference HF). • Consider the mode of transmission of the potential outbreak pathogen to define the essential quantities of PPE and supplies required. • Additional details can be obtained in the WHO disease commodity packages for technical specifications.	• G_IPC outbreak essential supplies list • WHO diseases commodity packages • Technical specifications of personal protective equipment for COVID-19 • WHO preferred product characteristics for personal protective equipment for the health worker on the frontline responding to viral hemorrhagic fevers in tropical climates
	Develop a coordination strategy with the logistics/supply management team.	• Establish a work group to define the strategy for purchasing, stockpiling and distributing supplies in coordination with the logistics/supply team. • Use the WHO technical specifications as a reference.	
	Develop plans for stockpiling IPC supplies and estimating burn rates of supplies.	• Develop an inventory of existing PPE stocks, considering the reference HFs. • Define a frequency or other criteria for timely replacement of PPE inventory. • Develop contingency plans in case of supply shortages and de-escalation of IPC supplies distribution. • The *PPE burn rate calculator* (CDC, Atlanta, US) can be used as a reference. • Adaptt surge planning support tool can be used.	• PPE burn rate calculator • WHO COVID-19 essential supplies forecasting tool (COVID-ESFT) v4.1 • Adaptt tool
	Create budget lines for IPC supplies.	• Consider the reference HF inventory of PPE and IPC supplies to develop preliminary budget lines.	• H_IPC outbreak budget template
	Develop simulation exercises addressing the IPC component of the outbreak preparedness plan.	• The WHO COVID-19 simulation exercises package can be used and adapted for other threats. • *Country example number 1* can be used as a reference.	• Health facility and infection prevention and control tabletop exercise for COVID-19

Key action	Steps	How to do	List of tools/resources
Develop national IPC outbreak guidelines	Develop or adapt guidelines from available materials on IPC outbreak management.	• Ensure you have a basic set of IPC guidelines: (a) standard precautions; b) transmission-based precautions; c) aseptic technique of invasive procedures; d) specific guidelines for most prevalent HAIs; e) environmental cleaning and disinfection; f) decontamination and reprocessing of medical equipment; g) screening, triage and isolation. • Obtain approval of the Ministry of Health (Ministry of Health) or equivalent authority. • Consult the WHE IPC & WASH for disease-specific IPC guidelines	• Standard precautions • Transmission-based precautions • Decontamination and reprocessing of medical devices for health-care facilities • Respiratory and hand hygiene
	Develop SOPs for IPC procedures.	• Define priority SOPs and assessment tools to be developed or adapted. • Use *aides-memoires* to reinforce the main concepts. • Interact with members of the clinical management team and nursing department to ensure the SOPs are embedded in the clinical procedures.	• F_IPC outbreak SOP template • General instructions for surface cleaning and disinfecting in patient care areas • How to make soapy water solution • Infection prevention and control (IPC) and water, sanitation and hygiene (WASH) in health emergencies
Develop national IPC outbreak training programme	Evaluate existing national IPC training strategy.	• Identify gaps as well the local available knowledge and expertise related to IPC in the context of outbreak response.	
	Map and/or evaluate IPC training programmes provided by other organizations.	• Identify potential partners among academic organizations, NGOs, public health agencies and funding agencies. • *Country example number 2* can be used as a reference.	• I_IPC outbreak partners mapping template • J_IPC outbreak training of trainers prototype • K_IPC outbreak training programme template
	Define a training coordination team for delivering the outbreak training programme.	• Consider collaborating with public health institutes and departments of epidemiology and disease control.	
	Define the outbreak training programme.	• Describe the targeted audience, learning objectives, teaching strategy, methods of evaluation. • Define the training strategy and ensure personnel with competence and training skills master the programme. • Consider using a *training-of-trainers strategy*. • *Country example number 2* can be used as reference.	
	Develop a basic training package for outbreak response	• Develop a practical training approach for essential IPC practices (hands-on training in clinical practice, or epi investigation) • Use as reference the *OpenWHO courses.*	• L_IPC outbreak training minimal content list • IPC course series
	Develop/adapt an IPC outbreak training programme for the rapid response team (RRT).	• Identify training needs related to IPC with the RRT team.	• WHO rapid response teams training
	Develop and maintain updated a database of trained personnel.	• Use the template of *training database.*	• M_IPC outbreak training database template
	Develop a system to evaluate the outcomes of training programme.	• Define a basic set of questions relevant for the training programme. Use a system of pre- and post-tests to assess the participants' learning.	

Key action	Steps	How to do	List of tools/resources
Coordinate with national surveillance and reporting programme	Evaluate the existing national or subnational IPC surveillance and reporting systems.	• Coordinate this action with the national or subnational surveillance networks to identify potential gaps related to outbreak management, with focus on inpatients and health and care workers. • *The WHO outbreak toolboxes* can be used as a reference about the disease on the focus.	• N_IPC outbreak surveillance checklist • WHO outbreak toolboxes
	Evaluate the available resources for laboratory, especially microbiology, capacity.	• Coordinate with other relevant national teams (e.g. surveillance, reference laboratory) to identify the capacity at national level.	
	Establish or reinforce the HAI surveillance strategy to timely identify and manage outbreak events.	• Establish a multidisciplinary technical working group to monitor IPC indicators. • Write down the HAI surveillance strategy element in the IPC component of the contingency plan.	
	Establish surge capacity for surveillance and reporting in a large-scale event.	• Ensure that the IPC focal points in reference/large HFs are trained in basic HAI surveillance procedures. • Define instructional material for surveillance that can be easily accessed by the IPC teams/focal points in HFs. • Develop a training programme for surveillance of outbreak-prone diseases. • Develop a roster of professionals who can be called upon to meet the needs for training. • The OpenWHO courses on *health-care associated infection* surveillance can be used as reference.	• M_IPC outbreak training database template • Health-care associated infection surveillance • Outbreak investigations in health facilities • Syndromic screening for infection prevention and control measures during public health emergencies
Develop national or subnational IPC communication strategy	Develop a communication strategy to disseminate IPC information.	• Develop a communication strategy to disseminate IPC information to various groups. (HFs, health and care workers, community and public, social media). • Identify designated spokesperson(s) at national or subnational levels. • *The WHO Risk communications and community engagement (RCCE)* tool can be used as reference.	• Communicating risk in public health emergencies • Communication for behavioral impact (COMBI) - toolkit
	Develop and verify standard communication materials.	• Specific outbreak-related WHO guidance references can be accessed on the webpage of *Health Emergencies – IPC and RCCE*.	• Health emergencies – infection prevention and control • Risk communication and community engagement (RCCE)
	Establish a work collaboration between risk communication and community engagement partners	• Identify the groups working on risk communication and community engagement at national level. • Identify relevant leaders in civil society who can be part of communication channels. • Identify scalable messaging channels that can be easily activated in an outbreak (e.g. HFs' e-mail addresses that can be used to deliver immediate information, social media).	• J_IPC outbreak partners mapping template

Key action	Steps	How to do	List of tools/resources
Develop surge capacity plans for IPC resources	Map HF and resources available.	• List the HFs that can receive suspected or confirmed cases of the disease at a threat. Identify key information, such as locality, number of beds available for intensive care and isolation rooms, and contact details of the IPC focal point. *The template for HF database* can be adapted to map the available resources.	• O_IPC outbreak HF database template • WHO -IPC in health facilities in the event of a surge or resurgence in cases of COVID-19
	Map partners and their respective IPC resources.	• Establish a list of potential partners and define a strategy for approaching and engaging with them. *The template for partners mapping* can be used.	• I_IPC outbreak partners mapping template
	Estimate and plan for increased IPC personnel requirements during outbreak situations.	• Adapt from the WHO Adaptt surge planning support tool.	• Adapt surge planning support tool
	Estimate budget items for IPC aspects of response.	• A basic budget spreadsheet can be adapted according to the needs.	• H_IPC outbreak budget template

B. Early priority: identify key personnel and partners

The early actions are the next set of actions required, subject to local conditions and feasibility. The focus is on identifying key staff and partners.

Table 2.3. Key actions, steps, how to do and list of tools resources for the preparedness public health emergency phase early priority.

Preparedness			
Early priority: identify key personnel and partners			
Key action	**Steps**	**How to do**	**List of tools/resources**
IPC outbreak task force	Establish an IPC outbreak task force, IPC committee, or IPC working group.	• Create a clear line of command to avoid duplication of efforts. If a national IPC committee already exists, this task force can be a derivative of it, although with different roles and responsibilities.	• P_IPC outbreak task force ToR prototype
	Establish mechanisms to meet regularly, record meetings, and identify decision-making processes.	• *The template for meeting minutes* can be used.	• R_IPC outbreak meeting minutes template
	Create a structure to cascade and disseminate information from the national level to facility level, and vice-versa.	• Define a procedure to deliver guidance and information.	• O_IPC outbreak HF database template

Key action	Steps	How to do	List of tools/resources
Establish IPC partner mapping and networking	Review and update the partner mapping.	• Update the partner mapping at least every six months, tracking its history of assistance. • Define the frequency of meetings.	• I_IPC outbreak partners mapping template
	Establish and maintain lines of communication.	• Establish regular forums for discussions to update the status of IPC implementation (at least every six months). Define the procedure to deliver guidance and information.	

C. Advanced priority: audit and test the system

The set of advanced actions is the final set of actions to be initiated, once immediate and early actions are under way. The focus is on auditing and testing the system.

Table 2.4. Key actions, steps, how to do and list of tools resources for the preparedness public health emergency phase advanced priority.

Preparedness			
Advanced priority: audit and test the system			
Key action	**Steps**	**How to do**	**List of tools/resources**
Assess IPC outbreak preparedness	Evaluate the status of IPC outbreak preparedness after immediate and early actions have been initiated.	• Use *IPC outbreak preparedness, readiness and response checklist* to Identify any gaps or deficiencies in the performance.	• R_IPC outbreak PRR checklist
	Define an action plan to improve areas with gaps.	• Undertake simulation exercise (e.g. tabletop exercises) to test the system. • *Country example number 1* can be used as reference.	• A_IPC outbreak action plan template • S_IPC outbreak simulation prototype

Chapter 3

Phase 2: IPC outbreak operational readiness

Chapter 3. Phase two: IPC outbreak operational readiness

3.1 Introduction

If the answer to Question 1 was "no" and the answer to Question 2 was "yes," then the country is in the readiness phase.

The emergency operational readiness phase links effective preparedness to efficient response and represents a statement of the capacity and capability of a relief agency or service. These activities may take up to six months to ensure readiness to respond to a specific, defined threat. *(1)* During this phase, resources developed during the preparedness phase should be activated – which suggests adopting and/or adjusting plans to target the threat at hand.

A summary of priority IPC key actions is provided in Table 3.1.

Table 3.1. Summary of key actions, priorities and respective multimodal strategy elements in the IPC operational readiness phase

Emergency management phase	Priority	Key actions	Multimodal element
Readiness	Immediate	Establish IPC outbreak taskforce	Build it
	"	Ready the national IPC outbreak response plan	Build it
	"	Adapt surge capacity plans for IPC resources	Build it
	"	Adapt national IPC outbreak training programme	Teach it
	"	Coordinate with national surveillance and reporting programme	Check it
	"	Adapt national or subnational IPC communication strategy	Sell it
	Early	Prepare IPC national partners	Live it
	"	Prepare IPC stakeholder networks	Live it
	Advanced	Assess IPC outbreak readiness	Check it

3.2 Step-by-step list of key actions and available tools for IPC operational readiness

A. Immediate priority: adapt existing IPC tools for outbreak operational readiness

The set of immediate actions is the first set of actions required in the phase. The focus is on operationalizing key activities, outlining roles and responsibilities, and identifying the necessary resources.

Table 3.2. Key actions, steps, how to do and list of tools resources for the operational readiness public health emergency phase immediate priority.

Operational Readiness			
Immediate priority: Adapt existing IPC tools for outbreak operational readiness			
Key action	**Steps**	**How to do**	**List of tools/resources**
Establish IPC outbreak task force	Either evaluate existing IPC outbreak task force, including reviewing the terms of reference, or, if a task force is not present, establish one.	• Revise and adapt the IPC outbreak task force terms of reference (ToR). • If this task force does not exist, create one. • *Case study number 1 and Country example number 2* can be used as reference.	• O_IPC outbreak task force ToR prototype
	Develop an action plan for the readiness phase.	• Describe the first steps to be taken, by whom and when.	• A_IPC outbreak action plan template
Ready the national emergency preparedness and response plan	Evaluate/review any existing IPC component of the national emergency contingency plan.	• Review or adapt according to the threat at hand. Define priority areas to be tackled based on the mode(s) of transmission and epidemiological context.	• E_IPC outbreak management plan template
	Review and adapt checklists, SOPs and assessment tools.	• Define the priority checklist, SOPs and assessment tools according to the mode of transmission of diseases. The priority forms are: • Develop or adapt from existing rapid HF IPC capacity assessment tools. • Case investigation and report for health care-associated infections. • Risk assessment and management for exposed health-care workers.	• F_IPC outbreak SOP template • Rapid hospital readiness checklist for COVID-19 • Infection prevention and control health-care facility response for COVDI-19 • Risk assessment and management of exposure of health care workers in the context of COVID-19: interim guidance, 19March2020 • Healthcare-associated Infection (HAI) outbreak investigation toolkit
	Define a rapid dissemination strategy of the developed/revised material.	• Establish communication lines with partners and HFs.	• I_IPC outbreak partners mapping template • O_IPC outbreak HF database template

Key action	Steps	How to do	List of tools/resources
	Coordinate IPC national command structures with other ministries or structures, minimizing any duplication of efforts.	• Establish a coordination strategy. • Define roles and responsibilities for IPC members of rapid response teams (RRT).	• WHO rapid response teams training
	Utilize drills, simulations or tabletop exercises to test the national IPC component of the emergency's preparedness and response plan.	• Adaptation can be done from other available threats. • *Country example number 1* can be used as a reference.	• Health facility and infection prevention and control table-top exercise
	Review any action budget items for IPC aspects of response.	• A basic budget spreadsheet can be adapted according to the needs. • If not already available, ensure budget allocation for IPC line items.	• R_IPC outbreak budget template
Adapt surge capacity plans for IPC resources	Review or map the priority HF and address any gaps.	• Define criteria for prioritizing health facilities. A basic HF spreadsheet can be adapted to create a database template. • Undertake rapid HF IPC capacity assessments and develop plans to address gaps.	• O_IPC outbreak HF database template • Rapid hospital readiness checklist for COVID-19 • IPC HF response for COVID-19 assessment
	Review/map the HF IPC personnel and their contact information.	• Update or develop a database or list of personnel trained in IPC.	• M_ IPC outbreak training database template
	Review and update a national stockpile of IPC supplies and equipment; if one is not available, create one.	• Anticipate PPE burn rates; identify and secure more funding according to the needs. • *Country example number 3* can be used as reference.	• G_IPC outbreak essential supplies list • PPE burn rate calculator • COVID-19 essential forecasting tool (COVID-ESFT) • R_IPC outbreak budget template
	Develop contingency plans for IPC resources or stockpiles.	• Determine alternate service delivery models to be used to conserve IPC supplies of PPE and to reduce the burden on the health system. • Develop IPC plans for handling larger number of deaths.	• E_IPC outbreak contingency plan list
	Create budget lines for IPC supplies.	• Consider the reference HF inventory of PPE and IPC supplies to develop preliminary budget lines.	• H_IPC outbreak budget template
Adapt national IPC outbreak training programmes	Activate a training coordination team for delivering the training strategy; if one is not already formed, create one.	• Collaborate with public health institutes, disease-control departments and other potential partners. • Review the mapping of IPC training programmes run by other organizations, including NGOs. • Implement or develop and implement the training strategy.	• I_IPC outbreak partners mapping template
	Evaluate existing national IPC training strategies and identify any gaps that need to be addressed to respond to the threat at hand.	• Implement a Training-of-Trainers (ToT) strategy. • *Country example numbers 2, 3 and 4* can be used as references. • Identify gaps as well the local available knowledge and expertise on the threat at a hand.	• J_ IPC outbreak training of trainers prototype • M_ IPC outbreak training database template

Key action	Steps	How to do	List of tools/resources
	Take IPC refresher courses and updates.	• Basic IPC training available at OpenWHO can be used as a reference.	• Infection prevention & control course series
	Develop a practical training approach for essential IPC practices (hands-on training in clinical practice, or epi investigation).	• The list of *IPC outbreak training minimal content list* can be used as a reference.	• K_IPC outbreak training programme template • L_IPC outbreak training minimal content list
	Implement the RRT training programme.	• With the RRT, identify training needs and take training related to IPC.	• WHO rapid response teams training
Coordinate with national surveillance reporting programme	Develop training evaluation.	• A questionnaire-based evaluation on the main topics of the list of *IPC outbreak training minimal content list.*	• L_IPC Outbreak training minimal content list
	Review existing national and subnational IPC surveillance and reporting systems.	• Activate or, if one is not available, establish a multidisciplinary technical working group to monitor IPC-relevant indicators. The IPC outbreak surveillance check list can be used as reference.	• N_IPC outbreak surveillance checklist • Syndromic screening for infection prevention and control measures during public health emergencies
	Develop IPC indicators for outbreak readiness and response monitoring.	• Examples of indicators: health-care workers' with infections; percentage of health-care workers who were trained; availability of IPC supplies.	• O_IPC outbreak HF database template • M_IPC outbreak training database template
	Establish a regular reporting mechanism that, includes epidemiological data for relevant stakeholders.	• Define the frequency of reports/meetings and outline of the standard content of the feedback.	
Adopt national or subnational IPC communication strategy	Evaluate the existing national or subnational IPC communication strategy and adapt to the threat at hand.	• In collaboration with risk-communication and community-engagement partners, evaluate the IPC communication strategy and develop a communication plan to respond to the threat at hand by defining modes of information deliverables, frequency, and content for health facilities. • Develop public messages directed to patients and visitors based on the threat at hand. • Ensure language used is appropriate for the audience being targeted.	• Communicating risk in public health emergencies • Communication for behavioral impact (COMBI) - toolkit
	Implement the IPC operational readiness communication plan.	• Use all media available – radio, TV, social media, etc. – to disseminate communication plans.	

B. Early priority: prepare key personnel and partnerships

The set of early actions is the next set of actions required, subject to local conditions and feasibility. The focus is on identifying key staff and partners.

Table 3.3. Key actions, steps, how to do and list of tools resources for the operational readiness public health emergency phase early priority.

Operational Readiness			
Early priority: prepare key personnel and partnerships			
Key action	**Steps**	**How to do**	**List of tools/resources**
Prepare IPC stakeholder networks in HFs	Evaluate the status of IPC partners mapping and international coordination.	• Review and update the partners mapping and plans for surge capacity.	• I. IPC outbreak partners mapping template
	Define lines of communication.	• Establish regular forums for discussions to update the status of IPC implementation. • Review coordination and roles.	
Prepare and coordinate national IPC partners	Evaluate the status of networks of IPC stakeholders at the HF level and address any gaps identified.	• Review/update the stakeholders mapping. Allocate partners for HFs where no support is available. • *Country example number 2* can be used as reference.	• D. IPC outbreak stakeholders mapping template
	Establish lines of communication.	• Establish modes of communication with HFs and community. • Test communications systems.	

C. Advanced priority: audit and test the system

The set of advanced actions is the final set of actions to be initiated, once immediate and early actions are under way. The focus is on auditing and testing of the system.

Table 3.4. Key actions, steps, how to do and list of tools resources for the operational readiness public health emergency phase advanced priority.

Operational Readiness			
Advanced priority: audit and test the system			
Key action	**Steps**	**How to do**	**List of tools/resources**
Assess IPC outbreak readiness	Evaluate the status of IPC outbreak operational readiness after immediate and early actions have been initiated.	• Assess the gaps in the operational readiness • Assess the status of IPC practices in HFs. • Identify a priority HF to perform an IPC practices assessment.	• R_IPC outbreak PRR checklist
	Define and implement an action plan to improve any gaps identified.	• Use the *action plan template* to drive and document the process. • Undertake simulation exercises (e.g. tabletop exercises) to test the system. • *Country example number 1* can be used as reference.	• A_IPC outbreak action plan template • Health facility and infection prevention and control tabletop exercise for COVID-19 • S_IPC outbreak simulation prototype

Chapter 4

Phase 3: IPC outbreak response

Chapter 4. Phase 3: IPC outbreak response

4.1 Introduction

If the answer to Question 1 was "yes", then the country is probably in the response phase.

Response is the stage in which emergency actions exceed the usual level of activities in response to a defined public health threat *(1)*. During this phase, many resources developed in the preparedness and readiness phases should be activated – that means that the plan should be transformed into concrete actions, or adapted to respond to the threat at hand.

A summary of key actions to take according to the priorities is provided in Table 4.1.

Table 4.1. Summary of key actions in the IPC response phase, priorities and respective multimodal element

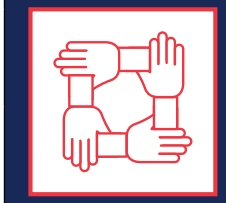
Emergency management phase	Priority	Key actions	Multimodal element
Response	Immediate	Activate IPC outbreak task force	Build it
	"	Activate national IPC outbreak response plan	Build it
	"	Activate surge capacity plans for IPC resources	Build it
	"	Leverage national IPC outbreak training programme	Teach it
	"	Activate national surveillance and reporting programme	Check it
	"	Activate national or subnational IPC communication strategy	Sell it
	Early	Monitor for resurgence of cases	Check it
	Advanced	Assess IPC outbreak response	Check it

4.2 Step-by-step list of key actions and available tools for IPC response

A. Immediate priority: activate existing and adapted IPC tools for outbreak response

The set of immediate actions is the first set of actions required in the response phase. The focus is on operationalizing key activities; outlining roles and responsibilities; and identifying the necessary resources. Once an outbreak has been declared and the incident management system (IMS) activated, convene the IPC task force.

Table 4.2. Key actions, steps, how to do and list of tools resources for the operational readiness public health emergency phase immediate priority.

Response			
Immediate priority: activate existing/adapted IPC tools for outbreak response			
Key action	**Steps**	**How to do**	**List of tools/resources**
Activate IPC outbreak task force 	Convene the IPC outbreak task force and adapt the terms of reference.	• Coordinate the IPC outbreak task force to revise and adapt the ToRs. • Define IPC task force meeting frequency and documentation of activities. • *Case study number 1 can be used as reference.*	• P_IPC outbreak task force ToR Prototype • Q_IPC outbreak meeting minutes template
	Develop an IPC response strategy and action plan.	• Focus on revising/adapting policies to specific threat at hand. • Review/adapt the IPC budget, including IPC supplies and waste management.	• A_IPC outbreak action plan template • H_IPC outbreak budget template
	Activate IPC stakeholders' networks.	• Update the partners' and stakeholders' mapping template, if necessary. • Define a rapid dissemination strategy for the developed/revised material.	• D_IPC outbreak stakeholders' mapping template • I_IPC outbreak partners' mapping template
Activate the national IPC outbreak preparedness and response plan	Activate the IPC components of any existing national or subnational outbreak response plan.	• Identify priority IPC areas to strengthen based on the threat at hand. • Activate outbreak response financing plans.	• E_IPC outbreak contingency plan list • G_IPC outbreak essential supplies list • H_IPC outbreak budget template
	Adapt or review policies/SOPs for the specific threat at hand.	• Adapt/update IPC guidance based on what is known about modes of transmission, incubation period, immunization and population at risk. Priorities: - Patient placement, transportation and referral; - Contact tracing and quarantine (if applicable); - Visitor management and crowd control in HFs; - Testing, exposure management and isolation of health and care workers (if applicable); - Use of PPE and other IPC supplies.	• F_IPC outbreak SOP template • WHO diseases outbreak toolboxes

Key action	Steps	How to do	List of tools/resources
Activate surge capacity plans for IPC resources	Review/map the HF and resource gaps.	• A basic HF spreadsheet can be adapted according to the needs. Identify national-level isolation/bed capacity (this may include establishing alternative isolation facilities) and, in collaboration with case management, develop a patient referral pathway.	• O_IPC outbreak HF database template
	Review/Map the IPC personnel and their contact information.	• Review or, if one is not available, develop a database or list of trained personnel. A basic training database spreadsheet can be adapted according to the needs. Allocate IPC personnel to priority HFs as needed.	• M_IPC outbreak training database template
	Activate or, if one is not available, establish a national stockpile of IPC supplies and equipment.	• Distribute or, if not already done, define essential IPC items to be provided to priority HFs. • Calculate PPE burn rates. • Work in cooperation with logistic team. • *Country example number 3* can be used as reference. • Ensure a plan is in place for HF whose personel have been repurposed for the response (e.g. reduce or discontinue certain services).	• G_IPC outbreak essential supplies list • PPE burn rate calculator • COVID-19 essential forecasting tool (COVID-ESFT)
	Develop contingency plans for IPC resources or stockpiles.	• Determine alternative service-delivery models to be used to conserve IPC supplies of PPE and to reduce the burden on the health system. • Develop IPC plans for handling larger number of deaths.	
Leverage and activate national IPC outbreak training programme	Evaluate existing national IPC training strategy and address any gaps for the event.	• Identify gaps as well the local available knowledge and expertise on the threat at a hand. • Undertake a training of trainers (ToT) or other strategy. • *Country example number 3* can be used as reference.	• I_IPC outbreak training trainers prototype
	Undertake refresher courses and updates.	• Update the educational material based on the threat at a hand.	• L_IPC outbreak training minimal content list • IPC training resources • OpenWHO IPC training courses catalog • OpenWHO IPC courses • WHO diseases outbreak toolboxes
	Review/map and evaluate IPC training programmes run by organizations, including NGOs.	• Collaborate with organizations, including NGOs, to ensure that training material is streamlined and consistent.	• I_IPC outbreak partners mapping template
Activate national surveillance reporting programmes	Activate/adapt existing national or subnational IPC surveillance and reporting systems appropriate for responding to the event at hand.	• The IPC outbreak surveillance check list can be used as reference. • If specific tools or policies are lacking for the specific threat at hand, develop or adapt them.	• N_ IPC outbreak surveillance checklist • Syndromic screening for infection prevention and control measures during public health emergencies
	Perform regular analyses and create reports on the current epidemiological situation.	• Use available data collected from outbreaks to respond to and inform IPC policies. • Review surveillance definitions as new data become available.	

Key action	Steps	How to do	List of tools/resources
Activate national or subnational IPC communication strategy	Activate specific IPC communication strategies for the outbreak at hand.	• Consult the existing national or subnational IPC communication strategy. • Activate IPC-specific components as needed for the outbreak at hand.	
	Disseminate IPC information to the targeted audience.	• Define and implement strategies according to the targeted audience (HFs, health and care workers, community) via various media (radio, TV, leaflets, social media, etc.). • Define modes of communication, frequency and content. • *Country example number 4* and *Case study number 2* can be used as references.	• Communicating risk in public health emergencies
	Ensure that messages are in appropriate languages(s) and that their content is relevant.	• Engage other disciplines, including behavioral and/or social sciences. • Engage with Risk Communication and Community Engagement (RCCE) team.	• Communication for behavioral impact (COMBI) - toolkit

B. Early priority: monitor for resurgence of cases

The set of early actions is the next set of actions required, subject to local epidemiology.

Table 4.3. Key actions, steps, how to do and list of tools resources for the operational readiness public health emergency phase early priority.

Response			
Early priority: monitor for resurgence of cases			
Key action	**Steps**	**How to do**	**List of tools/resources**
Monitor for resurgence of cases	Evaluate the epidemiological situation.	• Convene the IPC task force to discuss the results of the epidemiological situation related to cases in HFs and among health and care workers. If a resurgence of cases is identified, react as if in acute-response mode.	• A_IPC outbreak action plan template • Q_IPC outbreak meeting minutes template
	Once a resurgence is detected, review the steps in the Immediate Priority to identify improvements required.	• Revaluate IPC response strategy and action plan. • Revaluate surge capacity. • Update the training strategy and/or content. • Review and update surveillance definitions (if necessary). • Review the existing language and content of IPC messaging.	• A_IPC outbreak action plan template • E_IPC outbreak contingency plan list

C. Advanced priority: conduct intra-action or after-action review

The set of advanced actions is the final set of actions to be performed, once immediate and early actions are under way. The focus is on assessing the system.

Table 4.4. Key actions, steps, how to do and list of tools resources for the operational readiness public health emergency phase advanced priority.

Response			
Advanced priority: conduct intra-action or after-action review			
Key action	**Steps**	**How to do**	**List of tools/resources**
Assess IPC outbreak response	Evaluate IPC outbreak-response efforts.	• Define a review strategy, that includes both intra-action and after-action reviews to identify any gaps in the IPC-response plan. • Conduct a review of IPC response. • Develop a plan to address any gaps identified.	• A_IPC outbreak action plan template • R_IPC outbreak PRR checklist • WHO guidance for after action review

Chapter 5

Country examples and case studies

Chapter 5. Country examples and case studies

5.1 Introduction

This chapter provides practical examples of the key actions for the IPC outbreak preparedness, readiness, and response. Although these examples are not comprehensive, they may assist countries in adapting to their local context. Country examples were collected from IPC focal points in different countries; they focus on the barriers that the countries faced and describe how they overcome them. The case studies, which were identified in published literature, highlight outbreak response strategies to illustrate relevant key actions.

Box 5.1. Country example number 1.

Simulation for hospital preparation for ebola outbreak
Singapore

Barrier identified by the country: health facilities and systems potentially inadequately prepared and staff not trained to promptly recognize novel or emerging infectious diseases and take appropriate, rapid precautions.
Facilitator to overcome such a barrier: Coordination by Ministry of Health of a national-led simulation.

Phase:	**Readiness**
Priority:	**Immediate**
Key action:	**Ready the IPC outbreak response plan or Ready the national response plan**

Summary
International travel can introduce emerging infectious diseases with pandemic potential, making large cities worldwide highly vulnerable. Ideally, the health facilities in such cities should be prepared for any presentation of a patient with an infectious disease. In Singapore, all health facilities should be prepared for the presentation of a traveler with a novel, transmissible infectious disease. In 2014, the Ministry of Health Singapore recognized that the pending threat of secondary transmission of EBOD in travelers from West Africa needed to be addressed. To mitigate this threat, the Ministry of Health Singapore undertook a series of walks through the hospitals – from the emergency departments to the intensive care units (ICUs) – to assess their readiness. Each acute care hospital hosted such "walkabouts," which described patient journeys in various scenarios, including presentation; identification and early isolation; late identification on a ward; deterioration; and transfer to ICU, etc. Ministry of Health visitors could ask questions in order to better understand the hospital's capacities and risks.

To further test the facilities' readiness, the Ministry of Health undertook national simulation exercises. Three months prior to the simulation exercises, preparations commenced that included: a request from the Ministry of Health that each health facility provide a copy of its emergency SOP; Ministry of Health also set up a planning and evaluation team that included Ministry of Health staff and external experts. The team created a timeline for the scenarios so that it was clear when each of the steps would be taken and who would be involved[a]. Additionally, the team developed a comprehensive checklist[a1], which assessed the facilities response, infection prevention and control practices, contact tracing, communications, surveillance and epidemiology.

Ministry of Health staff rehearsed roles of the simulated patient with EBOD and the patient's relatives. The team identified exercise controllers, actors and evaluators to facilitate each exercise. Three scenarios were set out and used randomly in different health facilities on different dates. In order to ensure that health facilities would be able to continue normal operations during the simulation exercises, the exercise organizers gave health facilities three hours' notice before they got underway.

The first simulation exercise involved a patient with EBOD presenting to the Emergency Department who required transfer to the isolation ward. A second scenario involved a patient identified late as having EBOD in an open ward following a surgical procedure. The third scenario involved a child who was vomiting and required transfer to a different health facility.

Following each exercise, the team debriefed hospital staff, senior management and the evaluation team, which informed a written evaluation submitted by the Ministry of Health to the health facility. In cases where practices were identified as needing improvement, health facilities were required to submit plans to rectify them within one month after receiving the report.

The Ministry of Health concluded that its health facilities have robust emergency systems in place to identify and manage a a patient presenting with EBOD. The simulation exercise was found to be valuable in assessing a level of readiness for any infectious disease threat.

[a]The timeline and the checklist are available in the publication[1].

Source: This country example was taken from a summary of the publication below and following an interview with Dale Fisher. Used with permission.

Reference

1. Lum, L., Badaruddin, H., Salmon, S., Cutter, J., Lim, A and Fisher, D. Pandemic Preparedness: Nationally -Led Simulation to Test Hospital Systems. (2016) Annals Academy of Medicine. Vol 45: 8: 332-337.

Box 5.2. Country example number 2.

IPC national outbreak task force, training and stakeholders' network

Nigeria

Barriers identified by the country: poor ownership of IPC programmes by states; inadequate funding of IPC programmes; and lack of interest among some stakeholders.

Facilitators to overcome such a barrier: coordination of National IPC Programme by Nigeria Centre for Disease Control and Prevention (NCDC); convening of stakeholders (including states) to be involved in IPC work; sustaining the establishment of state-run IPC programmes; and advocating for better funding of IPC programmes.

Phase:	**Readiness**
Priority:	**Early**
Key action:	**Prepare IPC stakeholder networks**

Summary

Nigeria is a vast country with a population of over 200 million and a complex health system funded and regulated by federal and state governments. Many departments and agencies under the Federal Ministry of Health play important roles in infection prevention and control. The NCDC's role as the lead organization for the national IPC Programme is set out in the draft national IPC policy, awaiting endorsement by the Minister of Health. The National IPC Programme team members are consistently convening stakeholders to involve them in IPC work. These stakeholders include the Department of Hospital Services Federal Ministry of Health, the National Primary Health Care Development Agency, the Infection Control Africa Network (ICAN), the Dr Ameyo Stella Adadevoh Health Trust (DRASA), and academics from the Centre for Infection Control and Patient Safety of the College of Medicine, University of Lagos.

The IPC National Programme team members share national IPC plans with their stakeholders so that stakeholders can align their own organizations' plans and activities in a way to help achieve National IPC Programme objectives. As part of COVID-19 response activities, the National IPC Programme leading the IPC pillar of the Presidential Task Force on COVID-19, brought together various stakeholders, including partners (WHO, UNICEF (United Nations Children's Fund), USAID (United States Agency for International Development), police and civil defense and the Nigerian Army Medical Corps).

The IPC Programme is called by the Participants call the National IPC Programme "Turn Nigeria Orange". as a name Nigerians can relate to. The aim is for the color orange to describe to all stakeholders that they are turning Nigeria orange as they seek to strengthen IPC. The implementation strategy involves "Orange Networks," which are networks of health facilities that can interact as peers because they share certain characteristics. Those shared characteristics could be level of care, common regulators/funders or geographical proximity to one another). All IPC focal points in the health facility must be given the responsibility and authority to carry out their roles; they are responsible for developing action plans that are funded. Specifically, these networks are also involved in outbreak preparedness, readiness and response activities.

Phase:	**Response**
Priority:	**Immediate**
Key action:	**Establish/Activate the IPC outbreak task force**

Summary

The NCDC leads and coordinates the National IPC Programme, which has dedicated staff, a budget for the IPC programme, and close links to other aligned programmes, such as antimicrobial resistance, the Institute of Human Virology Nigeria, the AIDS Prevention Initiative, Patient Safety at the Department of Hospital Services Federal Ministry of Health and the National Primary Health Care Development Agency. At the national level, standing technical working groups routinely track epidemic-prone diseases such as Lassa fever, cholera, EBOD, cerebrospinal meningitis, COVID-19, and mpox. Each of these technical working groups has IPC pillars that provide leadership in the form of risk assessment, development of technical documents such as guidelines, standard operating procedures, workplace reminders, and IPC capacity (through the state and health facility IPC programmes) during the preparedness phases. The national IPC programme also makes inputs into the preparedness and response plans for the priority diseases.

The first COVID-19 case in Nigeria was confirmed on 28 February 2020. A few days later (17 March 2020), the Nigerian Government launched the Presidential Task Force (PTF) on COVID-19 to coordinate the country's multisectoral, intergovernmental response.[1] Each pillar was led by a different ministry, department, or agency. The IPC programme was part of this PTF, which focused on embedding and strengthening infection prevention and control programmes across the country. Among other activities, the task force assessed health facilities' IPC preparedness for dealing with COVID-19 cases across the country.[2] As part of the PTF's end-of-year review, it distributed to key stakeholders an online survey, which found that IPC practices had been enhanced and infections among health and care workers had been reduced.[1]

Phase:	Response
Priority:	Immediate
Key action:	Leverage national IPC outbreak training programme

Summary

Training is an essential part of ensuring that all health and care workers have the necessary skills to deliver safe health care and to prevent transmission of infections. Therefore, Nigeria requires that IPC professionals at national, state and health facility levels possess core competencies as outlined in the WHO core competencies for infection prevention and control professionals.[3] The national programme have also developed online training modules to support IPC professionals. Bespoke training-of-trainer sessions were set up early in the COVID-19 pandemic to ensure that all health and care workers were trained appropriately across the country. By December 2020, approximately 35 500 health and care workers had been trained on targeted IPC practices, including proper use of personal protective equipment.[1]

Source: This country example was a courtesy of Tochi Okwor, Coordinator of the Antimicrobial Resistance and Infection Prevention and Control Programme at the Nigeria Centre for Disease Control. Used with permission.

References

1. Bolu O, Mustapha B, Ihekweazu C, Muhammad M, Hassan A, Abdulwahab A, Asekun AA, Nsirim R, Okechukwu E, Attah I, Swaminathan M, Greby S, Adebiyi A, Alex-Okoh M, Okwor T, Ilori E, Mba N, Mutah J, Akujobi J, Battah N, Haggai W, Okatubo G, Okigbo A, Castle E, Abubakar I, Akataobi C, Adekunle O, Aliyu SH. Effect of Nigeria Presidential Task Force on COVID-19 Pandemic, Nigeria. Emerg Infect Dis. 2022 Dec;28(13):S168-S176. doi: 10.3201/eid2813.220254.

2. Joy Okwor T, Gatua J, Umeokonkwo CD, Abah S, Ike IF, Ogunniyi A, Ipadeola O, Attah T, Assad H, Dooga J, Olayinka A, Abubakar J, Oladejo J, Aderinola O, Eneh C, Ilori E, Ibekwe P, Ochu C, Ihekweazu C. An assessment of infection prevention and control preparedness of healthcare facilities in Nigeria in the early phase of the COVID-19 pandemic (February-May 2020). J Infect Prev. 2022 May;23(3):101-107. doi: 10.1177/17571774211060418.

3. WHO. Core Competencies for infection prevention and control professionals. World Health Organization September 2020. Core competencies for infection prevention and control professionals (who.int).

Box 5.3. Country example number 3.

Surge capacity and training

Yemen

Barriers identified by the country: fragile health infrastructure and limited available human resources; lack of SOP, protocols and guidelines for IPC that are properly adapted to the context of Yemen; lack of adherence by health and care workers to the IPC measures.

Facilitators to overcome such a barrier: development of a response plan in cooperation with partners, including WHO, other United Nations agencies, funds and programmes; cross-cutting collaborations with other pillars in the response.

Phase:	**Response**
Priority:	**Immediate**
Key action:	**Adapt/Activate surge capacity plans for IPC resources**

Summary

During early phases of the COVID-19 pandemic, when the health infrastructure and available human resources in Yemen were already overstretched, the demand for their services increased rapidly due to the increased number of COVID-19 cases. Almost 50% of health facilities were affected by the ongoing conflict, with reduced operational capacity: HeRAMS Assessment[1] showed that 2,477 health facilities (or 49% out of the 5056 health facilities across the country) were either not functional or only partially functional due to damages, staff shortages, lack of medicines and medical supplies or limited access due to the prevailing security environment. Most of the hospitals in Yemen do not have on-site waste-disposal systems and 70% of the health facilities do not have adequate WASH services. Only 30% of the health facilities are connected to the public network, which uses water trucks as alternate water sources.

The Yemen National COVID-19 Preparedness and Response Plan was prepared with the support of WHO, other United Nations agencies, funds and programmes and partners working in Yemen with input from donors. It provided guidance on actions to be taken to ensure that the country is able (1) to detect, test, isolate and treat individuals who become infected by the coronavirus as well as (2) to slow and contain the spread of the virus in the community. Considering these challenges, WHO had developed a response plan that was initially composed of eight pillars, where IPC was the sixth pillar of the COVID-19 response[2]. The objective of the IPC pillar strategy was to prevent or limit COVID-19 virus transmission in health facilities and other settings such as households, schools, mosques and during mass gatherings. Pillar 6 interventions were crosscutting and relevant to all other pillars of the response plan[2].

The first activity conducted was a risk assessment of IPC capacity – including availability of triage and appropriately ventilated isolation rooms – at all levels of the health-care system. Based on this assessment, a referral pathway was defined in collaboration with case management capacity (pillar 7). Particular attention was given to ensuring compliance with basic IPC principles in the designated Isolation units, points of entry and laboratories. A technical working group on IPC and WASH was created to address and coordinate the WASH and IPC needs of the health facilities and communities. A resource-mapping tool was adapted for WASH and IPC needs in health facilities. This tool was organized according to the governorate, facility name and district, and included critical IPC/WASH information such as whether isolation units were present; whether they were functional; what their support status was (e.g. percentage of medical staff who were trained, delivery of material and equipment, availability of staff); what organization was supporting the facility; and information on solid waste management, hand hygiene, water and sanitation provision.

In addition, the tool has information on the costs estimated for WASH/IPC needs. This tool was shared with partners to identify the available resources in the country and to identify the needs.

To ensure availability of WASH (safe water provision) and the IPC supplies needed to implement the recommended IPC protocols (e.g. hand hygiene resources, PPE, environmental cleaning, and waste management), partners including WHO scaled up the IPC measures in the existing Diarrhoea Treatment Centres, given the increased risk for cholera patients in terms of comorbidity.

Applied WASH, waste management and IPC practices were crucial at that point, given their relevance to COVID-19. Environmental cleaning and safe excreta management (faeces and urine) were prioritized, along with hand hygiene services, availability of disinfectants and safe waste management. As part of the safe health-care waste management system, the segregation, collection and disposal of waste are essential. This was an area where huge efforts to improve this sector were needed. In coordination with the organizations that specialized in risk communication and the health and WASH clusters, IPC guidelines were disseminated and applied in the target facilities, particularly in facilities that had been identified for isolation and treatment of COVID 19 patients. This was primarily to prevent onward transmission to staff, fellow patients and visitors.

Initial stock of PPE was made available to support the process of triage, referral pathways and case management. Additional PPE was procured in coordination with UNICEF. Support from other UN organizations such as the International Organization for Migration, the United Nations Refugee Agency and UNICEF was requested to help prepare adequate space for triage using tents in camps for internally displaced persons and points of entry. These organizations worked closely with WHO for technical guidance on IPC, triage and referral protocols.

Phase:	**Response**
Priority:	**Immediate**
Key action:	**Leverage/implement national IPC outbreak training programme**

Summary

During an outbreak of cholera in Yemen in 2020 health and care workers were caught unprepared: they did not follow IPC measures and there was no national IPC programme or systematic monitoring.

Having learned from that experience, the government and partners initiated training and capacity-building activities as part of a number of actions that were taken to ensure that health staff working in isolation facilities and quarantine areas were familiar with IPC protocols and requirements and followed them. During the COVID-19 pandemic, 3,048 health professionals were trained to support not only IPC activities but also the clinical management of COVID-19 patients. A training of trainers was used in the initial response; further, the trainers from Ministry of Health were sent later to governorates, where the training was conducted face to face.

Source: This country example was taken from a summary of the publication below and following an interview/communication with Nosheen Usman, WHO WASH and IPC focal point in Yemen.

References

1. Health Resources and Services Availability Monitoring System (HeRAMS) (https://herams.org/session/create, accessed 14 February 2023).

2. Yemen COVID-19 response Project (https://www.emro.who.int/yemen/information-resources/yemen-covid-19-response-project.html#:~:text=The%20Yemen%20COVID%2D19%20Response,systems%20for%20public%20health%20preparedness, accessed 14 February 2023).

Box 5.4. Country example number 4.

IPC large scale, on-the-job training and communication strategy during the outbreak of COVID-19

Iraq

Barriers identified by the country: challenges due the lack of information about the disease in the early stages of the COVID-19 pandemic; increasing numbers of cases in a short time period, as well as the lockdown.

Facilitators to overcome such barriers: the use of WHO guidelines for acute respiratory infections in health care and WHO interim guidance on the novel coronavirus; the engagement of the IPC medical city focal point; and support from the WHO regional focal point.

Phase:	Readiness or Response
Priority:	Immediate
Key action:	Adapt/leverage of national IPC outbreak training programme

Summary

Iraq created its national IPC programme in 2006; however, it had no dedicated budget and was not very active.

In January 2020, motivated by the spread of COVID-19 in neighboring countries, Iraq started its readiness and response plan for COVID-19. At the time, Iraq's health-care infrastructure was weak, the result of decades of conflict. The response plan included the designation of one COVID-19 facility per governorate[1]. By February 2020, the IPC focal point and the team in the Ministry of Health had to deal with the scarcity of reliable information that was then available to them. Using what information they did have along with WHO guidance on IPC, they planned a ToT programme.[2, 3] This programme was planned to be a three-day training course for up to 25 health and care workers of different backgrounds (including cleaners). The goal was to minimize health care-associated infections and thereby protect patients, visitors and health and care workers.

However, as the epidemiological situation evolved, marked by a lockdown and increasing limitations on large-scale, in-person training programmes, the country's Ministry of Health IPC team changed its strategy to one that entailed two to three hours of on-the-job training for up to 15 health and care workers. To achieve a larger-scale training, the IPC medical city focal point supported by the WHO IPC regional focal point set up virtual training sessions. The focus of this on-the-job training was the hand hygiene, the standard precautions, and the use of PPE. The Ministry of Health team concluded that large-scale training was feasible during an epidemic through on-the-job training.

Phase:	Response
Priority:	Immediate
Key action:	Activate national or subnational IPC communication strategy

Summary

Due to the difficulties in reaching and updating staff members, and considering the rapid evolution of changes in the IPC recommendations, the IPC focal point in the Ministry of Health established an ongoing telephone mobile application group (via WhatsApp) with 170 focal points in health facilities. The objective of this group was – and remains – to update participants, through regularly scheduled technical updates as well as through rapid communications, whenever required. Participants have credited this system with facilitating effective

communication and improving overall productivity. However, some participants have noted that the system is not helpful in cases where an official letter or document may be necessary.

Source: This country example was taken from a summary of the publication below and following an interview/communication with Jamal A. Kadhim, IPC Programme Manager Iraq Ministry of Health/ Technical Affairs Directorate and Dina Essam (MRO, IPC external consultant).

References

1. Hossain SMM, Al-Dahir S, Hilfi RA, Majeed Y, Rahi A, Sabaratnam V, Al-Mulla T, Hossain O, Aldahir A, Norton R, Lami F. Evaluation of dedicated COVID-19 hospitals in the pandemic response in Iraq: pandemic preparation within a recovering healthcare infrastructure. BMJ Glob Health. 2022 Jun;7(Suppl 3):e008715. doi: 10.1136/bmjgh-2022-008715. PMID: 35750345; PMCID: PMC9226466.

2. World Health Organization. Infection prevention and control of epidemic- and pandemic-prone acute respiratory infections in health care. WHO: Geneva; 2014 (https://iris.who.int/handle/10665/112656, accessed 14 February 2023).

3. World Health Organization. (2020). Infection prevention and control during health care when novel coronavirus (nCoV) infection is suspected: interim guidance. WHO; 10 January 2020 (https://apps.who.int/iris/handle/10665/332447, accessed 14 February 2023).

Box 5.5. Case study number 1.

Outbreak of *Clostridioides difficile* infection in acute care hospitals

Pathogen/context: outbreak of a new strain of *Clostridioides difficile* infection (CDI) PCR rybotype 027 in three acute care hospitals. Country: the United Kingdom (Northern Ireland), 2007-2008.

This study demonstrates the importance of having a centralized team control the outbreak; the outbreak-control team can be considered here as a proxy for an IPC outbreak task force.

Phase:	**Response**
Priority:	**Immediate**
Key action:	**Activate IPC outbreak task force**

Summary

In 2007, an outbreak of 318 cases of *C. difficile* spread among three acute care hospitals of the same Trust (Northern Health and Social Care Trust in Northern Ireland). An outbreak control team – responsible for the management, investigation and control of the outbreak – was established in the trust, which developed an action plan addressing communication, infection control practices, clinical management, antibiotic restriction policy, environmental hygiene and surveillance. In the hospitals, an infection control team was activated to monitor the use of the care bundle to reduce the risk from *C. difficile*. Additionally, Hospital Site Implementation Teams comprising a senior manager, ward nursing and medical staff, and hotel services staff were formed to ensure that the outbreak control team action plan was implemented. The findings in the study highlighted the value of antimicrobial stewardship as essential in controlling outbreaks of *C. difficile* infections.

Source: this example was taken from the publication of Aldeyab et al (2011)[1].

Reference

1. Aldeyab MA, Devine MJ, Flanagan P, Mannion M, Craig A, Scott MG, Harbarth S, Vernaz N, Davies E, Brazier JS, Smyth B, McElnay JC, Gilmore BF, Conlon G, Magee FA, Elhajji FW, Small S, Edwards C, Funston C, Kearney MP. Multihospital outbreak of Clostridium difficile ribotype 027 infection: epidemiology and analysis of control measures. Infect Control Hosp Epidemiol. 2011 Mar;32(3):210-9. doi: 10.1086/658333. PMID: 21460505.

Box 5.6. Case study number 2.

Global communication strategy during COVID-19 pandemic

Pathogen/context: COVID-19 pandemic; global health challenge when traditional modes of knowledge dissemination have not been feasible. Country: not specified (global initiative), 2020.

This example demonstrated that adapted virtual communities of practice and other learning networks might be valuable tools for rapid and large-scale communication and dissemination of information.

Phase:	**Response**
Priority:	**Immediate**
Key action:	**Activate national or subnational IPC communication strategy**

Summary

During the COVID-19 pandemic, there was a need for rapid dissemination of good-quality information, sharing of guidance and examples. The Extension for Community Healthcare Outcomes (Project ECHO) model was adapted to create a global network, which was designated as IPC Global Webinar Series. The objective was to bring together IPC professionals and experts on specific topics, aiming to facilitate the global response to COVID-19. Webinars were translated into five languages and recorded for public access. Professionals from an average of 100 countries participated in each of the 13 webinar sessions (May 2020 - August 2020). The number of participants in each session ranged from 393 to 1 181.

Source: this example was taken from the publication of Wilson et al, 2021.[1]

Reference

1. Wilson K, Dennison C, Struminger B, Armistad A, Osuka H, Montoya E, Padoveze MC, Arora S, Park B, Lessa FC. Building a Virtual Global Knowledge Network During the Coronavirus Disease 2019 Pandemic: The Infection Prevention and Control Global Webinar Series. Clin Infect Dis. 2021 Jul 15;73(Suppl 1):S98-S105. doi: 10.1093/cid/ciab320.

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Annex A.1

List of tools and resources

Tools

Box A1.1. List of tools for implementation of IPC-OFT.

Type	Code	Title
Template	A	IPC outbreak action plan
Template	B	IPC outbreak prioritization
Prototype	C	IPC outbreak prioritization workshop
Template	D	IPC outbreak stakeholders mapping
List	E	IPC outbreak contingency plan
Template	F	IPC outbreak standard operation procedure (SOP)
List	G	IPC outbreak essential supplies
Template	H	IPC outbreak budget template
Template	I	IPC outbreak partners mapping
Prototype	J	IPC outbreak training of trainers (ToT)
Template	K	IPC outbreak training programme
List	L	IPC outbreak training minimal content
Template	M	IPC outbreak training database
Checklist	N	IPC outbreak surveillance
Template	O	IPC outbreak HF database
Prototype	P	IPC outbreak task force terms of reference (ToR)
Template	Q	IPC outbreak meeting minutes
Checklist	R	IPC outbreak preparedness, readiness, response (PRR)
Prototype	S	IPC outbreak simulation exercise
Template	T	IPC & WASH HERO CAPE

Other resources

Box A1.2. List of other resources for implementation of IPC-OFT.

Type	Title	Available at
Assessment tool	Assessment tool of the minimum requirements for IPC programmes at national level	https://www.who.int/publications/m/item/assessment-tool-of-the-minimum-requirements-for-infection-prevention-and-control-programmes-at-the-national-level
Assessment tool	Infection prevention and control health-care facility response for COVID-19	https://www.who.int/publications/i/item/WHO-2019-nCoV-HCF_assessment-IPC-2020.1
Assessment tool	(IPCAF) WHO Infection prevention and control assessment framework at the facility level - IPCAT	https://www.who.int/publications/i/item/WHO-HIS-SDS-2018.9
Assessment tool	Instructions for the (IPCAT-2) National Infection Prevention and Control Assessment tool 2 (IPCAT-2)	https://apps.who.int/iris/handle/10665/330078
Assessment tool	component of States Paries Self-Assessment Annual Report (SPAR)	https://www.who.int/publications/i/item/9789240040120
Assessment tool	Joint External Evaluation tool: Internacional Health Regulation (2005) third edition	https://www.who.int/publications/i/item/9789240051980
Assessment tool	WASH FIT	https://www.washinhcf.org/wash-fit
Aide-memoire	Decontamination and reprocessing of medical devices for health facilities	https://www.who.int/publications/i/item/WHO-UHL-IHS-IPC-2022.4
Aide-memoire	IPC in health facilities in the event of a surge or resurgence in cases of COVID-19	https://www.who.int/publications/i/item/WHO-2019-nCoV-IPC-Surge-2021.1
Aide-memoire	Respiratory and hand hygiene	https://apps.who.int/iris/bitstream/handle/10665/341418/9789289055451-eng.pdf?sequence=1&isAllowed=y
Aide-memoire	Standard precautions for the prevention and control of infections	https://www.who.int/publications/i/item/WHO-UHL-IHS-IPC-2022.1

Type	Title	Available at
Aide-memoire	Transmission-based precautions for the prevention and control of infections	https://www.who.int/publications/i/item/WHO-UHL-IHS-IPC-2022.2
Assessment tool	Assessment tool on infection prevention and control minimum requirements for primary health care facilities	https://www.who.int/publications/i/item/WHO-UHL-IHS-IPC-2023.2
Assessment tool	Infection prevention and control and water, sanitation and hygiene in health facilities during Ebola or Marburg disease outbreaks: Infection prevention and control and water, sanitation and hygiene in health facilities during Ebola or Marburg disease outbreaks: rapid assessment tool user guide	https://www.who.int/publications/i/item/9789240107205
Assessment tool	Infection prevention and control and water sanitation and hygiene in health facilities during mpox disease outbreaks: rapid assessment tool, user guide	https://iris.who.int/items/df1ecf82-2ea7-41fd-b490-abcf727d0a60
Assessment tool	Joint External Evaluation Tool	https://www.who.int/publications/i/item/9789240051980
Check list	Rapid hospital readiness checklist for COVID-19	https://apps.who.int/iris/handle/10665/332778
Forecast tool	Personal protective equipment (PPE) burn rate calculator	https://www.cdc.gov/niosh/healthcare/hcp/pandemic/PPE-Burn-rate-calculator.html
Forecast tool	Adaptt Surge Planning Support Tool	https://www.who.int/europe/tools-and-toolkits/strengthening-the-health-system-response-to-covid-19/Surge-planning-tools
Forecast tool	WHO COVID-19 essential supplies forecasting tool (COVID-ESFT), version 4	https://apps.who.int/iris/handle/10665/340747
Guide	WHO guidance for contingency planning	https://www.washinhcf.org/wash-fit
Guide	WHO disease commodity packages	https://www.who.int/emergencies/disease-commodity-packages
Guide	WHO Technical specifications of personal protective equipment for COVID-19	https://www.who.int/publications/i/item/WHO-2019-nCoV-PPE_specifications-2020.1
Guide	WHO preferred product characteristics for personal protective equipment for the health worker on the frontline responding to viral haemorrhagaic fevers in tropical climates	https://www.who.int/publications/i/item/9789241514156
Guide	WHO Communicating risk in public health emergencies: a WHO guideline for emergency risk communication (ERC) policy and practice	https://www.who.int/publications/i/item/9789241550208
Guide	Risk assessment and management of exposure of health-care workers in the context of COVID-19: interim guidance, 19 March 2020	https://apps.who.int/iris/handle/10665/331496
Guide	WHO guidance for after-action review	https://apps.who.int/iris/bitstream/handle/10665/311537/WHO-WHE-CPI-2019.4-eng.pdf?sequence=1&isAllowed=y
Guide	Decontamination and reprocessing of medical devices for health-care facilities	https://www.who.int/publications/i/item/9789241549851

Type	Title	Available at
Guide	Infection prevention and control and water, sanitation and hygiene measures for diphtheria in health-care settings: operational guide	https://www.who.int/publications/i/item/9789240088160
Guide	How to make soapy water solution	https://www.who.int/multi-media/details/how-to-make-soapy-water-solution
Guide	Syndromic screening for infection prevention and control measures during public health emergencies	https://www.who.int/publications/i/item/B09222
Guide	General instructions for surface cleaning and disinfecting in patient care areas	https://www.who.int/publications/m/item/general-instructions-for-cleaning-and-disinfecting
Guide	Infection prevention and control guideline for Ebola and Marburg disease, August 2023	https://www.who.int/publications/i/item/WHO-WPE-CRS-HCR-2023.1
Guide	Infection prevention and control and water, sanitation and hygiene measures for Crimean-Congo haemorrhagic fever in health-care settings: operational guide	https://www.who.int/publications/i/item/9789240093188
Tabletop exercise	Health facility and infection prevention and control tabletop exercise for COVID-19	https://www.who.int/emergencies/diseases/novel-coronavirus-2019/training/simulation-exercise
Toolboxes	WHO disease outbreak toolboxes	https://www.who.int/emergencies/outbreak-toolkit/disease-outbreak-toolboxes
Tookit	Healthcare-associated infection (HAI) outbreak investigation	https://www.cdc.gov/healthcare-associated-infections/php/toolkit/outbreak-investigations-toolkit.html?CDC_AAref_Val=https://www.cdc.gov/hai/outbreaks/outbreaktoolkit.html
Toolkit	Communication for behavioural impact (COMBI)	https://www.who.int/publications/i/item/communication-for-behavioural-impact-(combi)
Training material	IPC course series	https://openwho.org/channels/ipc?locale=en
Training material	Rapid Response Teams Training Programme	https://extranet.who.int/hslp/package/rapid-response-teams-training-programme#:~:text=The%20%E2%80%9CRRT%20Training%20Programme%E2%80%9D%20(,at%20national%20and%20subnational%20levels.
Training material	IPC courses (COVID-19 and EBOD)	https://openwho.org/channel/Infection+prevention+and+control/521973
Various	Infection prevention and control (IPC) and water, sanitation and hygiene (WASH) in health emergencies (framework and guidelines ; general guidance and tools)	https://www.who.int/emergencies/operations/ipc-wash

Annex A.2

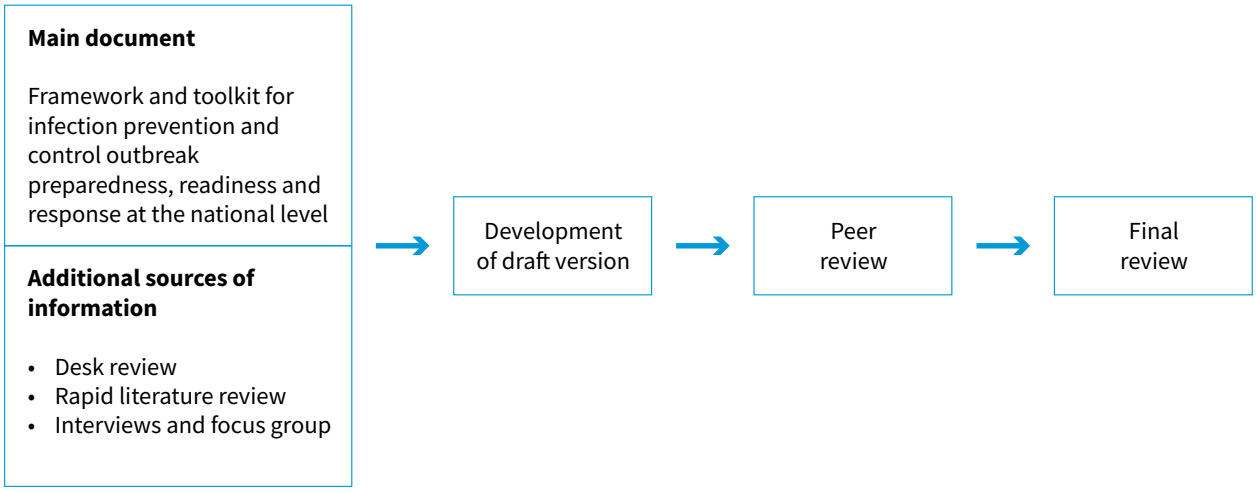
Methods

Methods

Design

WHO used a multi-pronged approach (see Figure A.2.1) to develop the implementation manual for the "Framework and toolkit for infection prevention and control outbreak preparedness, readiness and response at the national level (IPC-OFT)". It included additional sources of information and literature related to implementation concepts. *(1, 2)* During several meetings, the coordination team discussed the results of data collection and the structure of the implementation manual.

Figure A.2.1. Summary of steps in the development of the implementation manual.



Sources of information

In developing the manual, WHO drew primarily from the IPC-OFT. *(3)* Other sources included:

- Desk review: a broad search of published material from WHO and other international organizations on IPC implementation-related resources, including guidelines, trainings, assessment tools, checklists, etc.
- Rapid scoping literature review: to explore barriers to and enablers of IPC preparedness, readiness and response policies and interventions at national/subnational and health facility levels, as well to identify case reports containing descriptions of IPC practices and tools to manage outbreaks.
- Key stakeholder interviews and focus groups: to identify implementation barriers and facilitators through the experiences of IPC regional focal points and IPC national and subnational focal points.

Rapid scoping literature review

This review was intended to map the literature. The framework "population, concept, context (PCC) was used to determine the research question and eligibility criteria, as following:

- (P) Population (health workers, patients and visitors, government-level and Ministry of Health officials;
- (C) Concept (infection prevention and control programs and strategies, preparedness, readiness, response and outbreak); and
- (C) Context (health care).

Systematic reviews, other review types and primary papers were included, with no time limit, in English, Spanish and Portuguese. The following electronic resources were searched: PubMed, CINAHL, Cochrane Library, and Embase. Study authors were not contacted. A search strategy was developed for each database. Electronic reference manager Mendeley was used.

Papers were selected in two stages: one reviewer looked at selections of titles and abstracts and full text of selected citations; a second reviewer validated the selections through verification of a sample of papers (30%). Several meetings were held to reach a consensus on the selections.

Extracted data included details about authors, year, title, country, study objective, pathogen/disease, transmission-prevention interventions, infrastructure and workforce capacity, service effectiveness response and IPC intervention barriers. Case reports of IPC planning and practices and IPC tools were logged as well.

Interviews and focus groups

The WHO team developed a questionnaire for the interviewees and focus groups members to identify barriers to the IPC preparedness, readiness and response as well as to identify strategies that proved successful in overcoming those barriers. The participants were IPC experts, regional focal points and national focal points.

The coordination team developed a preliminary list of participants and expanded it through the snowball process. The team emailed invitations to 52 potential participants. Each interview took around one hour and was conducted through use of a virtual platform. In all, 20 people were invited to participate in a focus group, which lasted 90 minutes. The interviewees and focus groups participants were advised that the activity would be recorded for transcription purposes. They were told that their comments would be held in confidence except in cases where the team felt that it would be valuable to share a case study with other countries; in those cases, the participants' permission to divulge the comments would be sought.

Prior to the interview, several IPC focal points completed an online survey that assessed their level of awareness of and use of the IPC-OFT.

Summary of results

Rapid scoping literature review

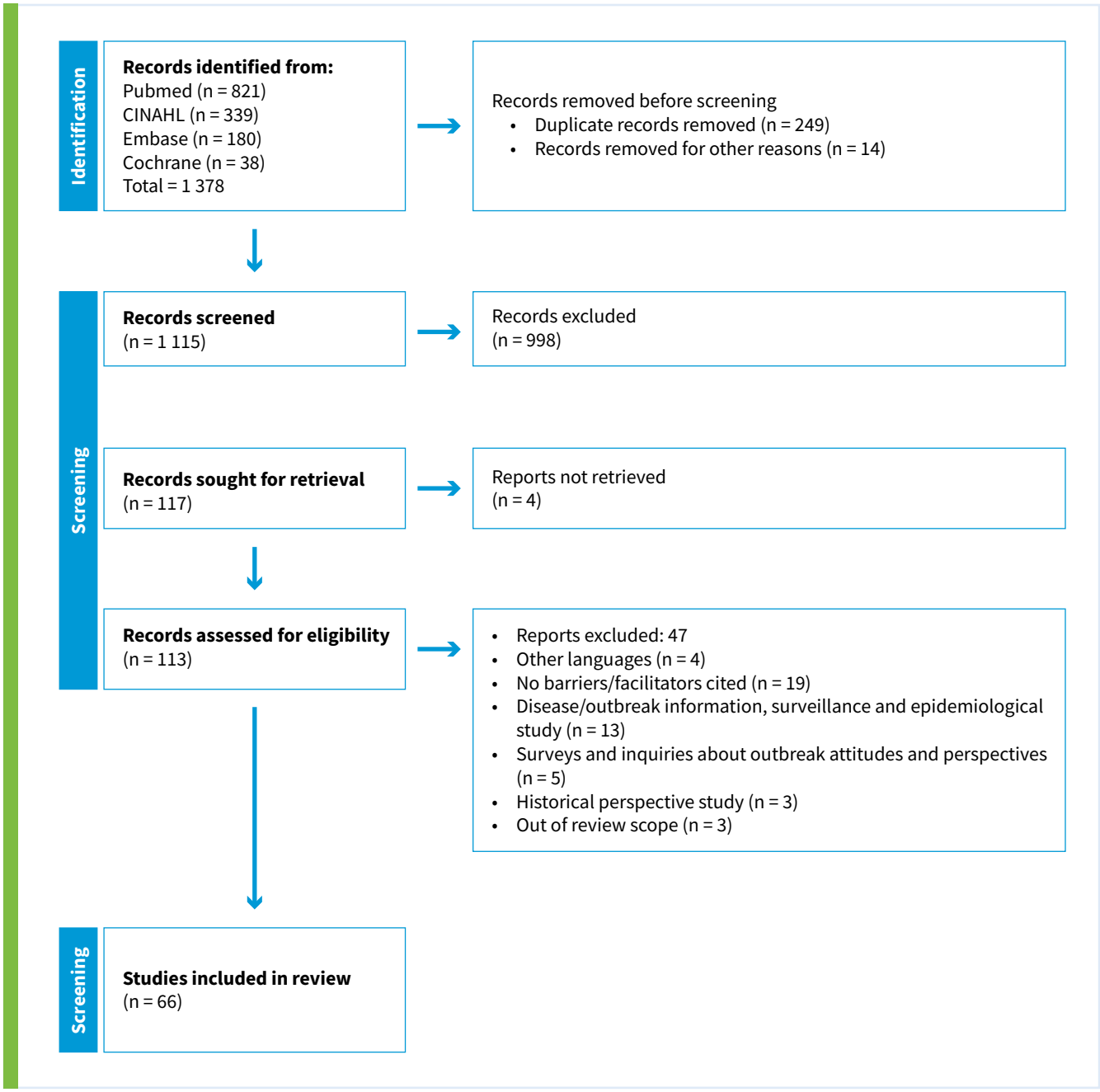
The search retrieved 1 378 materials, including 66 papers (Figure 2). Most of the studies were descriptive; six reviews were incorporated. The majority of the studies described developed practices and action plans. Fewer papers discussed national/subnational strategies. Studies from high-, low- and middle-income countries were incorporated in the synthesis. Primary- and tertiary-care levels were included; however, studies in hospital settings prevailed in this review. Facilitators of and barriers to IPC policies and interventions were synthesized into empirical categories.

In summary, the following categories were identified:

- Facilitators: transmission prevention interventions (health communications, physical distancing, monitoring and surveillance, health workers' training and information, protective measures for health and care workers, visits and patients' families' arrangements), infrastructure and supplies, workforce capacity, service effectiveness (systems, protocols and checklists, service planning, cases management, maintenance of essential services, team communication).
- Barriers: non-existent or ineffective IPC plans, health workers' education gaps, low compliance with IPC measures, lack of resources, laboratory capacity issues, differences between national and subnational levels' policies and practices.

Additionally, publications were selected that provided case studies to highlight outbreak-response strategies and to illustrate relevant key actions. Two of these cases are in Chapter 5 of the implementation manual.

Figure A.2.2. PRISMA flow diagram



Interviews and focus groups

During 2022, 15 interviews and three focus groups, with an average of six to eight participants in each group, were held. The records were transcribed and analyzed according to the thematic categories of barriers, namely:

- lack of awareness of the IPC-OFT and public health emergency terminology;
- lack of coordination;
- lack of IPC structure in health facilities;
- lack of support to health facilities from the national level;
- lack of IPC specialists; and
- inefficient provision of supplies.

The participants cited examples of strategies to overcome these barriers. These strategies were compiled with the rapid scoping literature review, and organized according to the multimodal strategy elements. The results are summarized in Table 1.2. of the implementation manual.

From the interviews and focus group activities, some country examples were identified and described in partnership with the contributors (see Chapter 5 of the implementation manual).

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